

Drosophila model to clarify the pathological significance of OPA1 in autosomal dominant optic atrophy

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Abstract

Autosomal dominant optic atrophy (DOA) is a progressive form of blindness caused by degeneration of retinal ganglion cells and their axons, mainly caused by mutations in the OPA1 mitochondrial dynamin like GTPase (*OPA1*) gene. *OPA1* encodes a dynamin-like GTPase present in the mitochondrial inner membrane. When associated with OPA1 mutations, DOA can present not only ocular symptoms but also multi-organ symptoms (DOA plus). DOA plus often results from point mutations in the GTPase domain, which are assumed to have dominant negative effects. However, the presence of mutations in the GTPase domain does not always result in DOA plus. Therefore, an experimental system to distinguish between DOA and DOA plus is needed. In this study, we found that loss-of-function mutations of the *dOPA1* gene in *Drosophila* can imitate the pathology of optic nerve degeneration observed in DOA. We successfully rescued this degeneration by expressing the human *OPA1* (*hOPA1*) gene, indicating that *hOPA1* is functionally interchangeable with *dOPA1* in the fly system. However, mutations previously identified did not ameliorate the *dOPA1* deficiency phenotype. By expressing both WT and DOA plus mutant *hOPA1* forms in the optic nerve of *dOPA1* mutants, we observed that DOA plus mutations suppressed the rescue, facilitating the distinction between loss-of-function and dominant negative mutations in *hOPA1*. This fly model aids in distinguishing DOA from DOA plus and guides initial *hOPA1* mutation treatment strategies.

eLife assessment

This study provides **valuable** insights into the complex genetics of dominant optic atrophy. Leveraging a fly model, the investigators provide **solid** evidence, albeit with small effect sizes, for a dominant negative mechanism of certain pathogenic variants that tend to cause more severe phenotypes, a long held hypothesis in the field. The work is of high interest to those in the optic atrophy and degeneration fields.

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Introduction

Autosomal dominant optic atrophy (DOA) is a progressive form of blindness characterized by selective degeneration of retinal ganglion cells (RGCs) and the axons that form the optic nerve. Despite being a rare disease with a frequency of 1/12,000 to 1/50,000, DOA is the most commonly diagnosed form of hereditary optic neuropathy. In 2000, *OPA1 mitochondrial dynamin like GTPase* (*OPA1*) was identified as the causative gene for DOA (Alexander et al., 2000 [↗](#); Deletrre et al., 2000 [↗](#)).

OPA1 encodes a dynamin-like GTPase located in the inner mitochondrial membrane (Olichon et al., 2002 [↗](#)). It has various functions, including mitochondrial fusion, mtDNA maintenance, control of cell death, and resistance to reactive oxygen species (ROS) (Lenaers et al., 2021 [↗](#)). The most common pathogenic mutation in *OPA1* is the c.2708-2711 delTTAG deletion (Toomes et al., 2001 [↗](#)). Five pathogenic mutations, including the one above, result in a substantial decrease in *OPA1* protein in the fibroblasts from affected patients (Zanna et al., 2008 [↗](#)), supporting the theory that DOA is caused by haploinsufficiency.

Optic atrophy is a characteristic feature of *OPA1*-associated DOA; however, multisystem symptoms have been reported in up to 20% of *OPA1* mutation carriers (Yu-Wai-Man et al., 2010a [↗](#)). This is called DOA plus. DOA plus can include sensorineural deafness (Amati-Bonneau et al., 2008 [↗](#)), multiple sclerosis-like illness (Verny et al., 2008 [↗](#); Yu-Wai-Man et al., 2016 [↗](#)), parkinsonism and dementia (Carelli et al., 2015 [↗](#); Lynch et al., 2017 [↗](#)), and cardiomyopathy (Spiegel et al., 2016 [↗](#)). The R445H mutation is a well-known mutation associated with DOA plus (Amati-Bonneau et al., 2008 [↗](#), 2003 [↗](#); Shimizu et al., 2003 [↗](#)). The proportion of mutated *OPA1* locations differs between patients with DOA and DOA plus. Missense mutations in the dynamin-featured structure, especially the GTPase domain of *OPA1*, are more likely to cause severe symptoms compared with loss-of-function (LOF) mutations such as deletions or splice site mutations (Yu-Wai-Man et al., 2010a [↗](#)). This is probably due to the dominant negative effect (DN) of a DOA plus mutation on the normal *OPA1* allele, since *OPA1* functions as a homo-oligomer (Frezza et al., 2006 [↗](#); Olichon et al., 2006); in fact, its yeast ortholog, Mgm1, forms oligomers, increasing the GTPase activity (Meglei and McQuibban, 2009 [↗](#); Rujiviphat et al., 2009 [↗](#)). Particularly, the dynamin-featured structure accounts for 70% of the 233 pathogenic variants of DOA and DOA plus described in the locus-specific database (Ferré et al., 2015 [↗](#)). However, mutations in these domains do not determine DOA plus, as it affects only 20% of all *OPA1* mutation carriers. Thus, experimental models to determine whether a gene mutation is LOF or DN are required to distinguish between DOA and DOA plus. Although currently there is no effective treatment for DOA or DOA plus, distinguishing between them allows earlier planning of interventions such as hearing aids, rather than relying solely on visual aids, necessary only for those with DOA plus. In the quest to differentiate between LOF and DN effects within the context of genetic mutations, precedents exist in simpler systems such as yeast and human fibroblasts. These models have provided valuable insights into the

conserved functions of OPA1 across species, as evidenced by studies in yeast models (Del Dotto et al., 2017) and fibroblasts derived from patients harboring *OPA1* mutations (Kane et al., 2017). However, the ability to distinguish between LOF and DN effects in an *in vivo* model organism, particularly at the structural level of retinal axon degeneration, has remained elusive. This gap underscores the necessity for a more complex model that not only facilitates molecular analysis but also enables the examination of structural changes in axons and mitochondria, akin to those observed in the actual disease state.

As *OPA1* is conserved in various species, DOA models have been reported in vertebrates such as mice and zebrafish, simple organisms such as nematodes and fruit flies, and *in vitro* yeast and cultured cell models (Del Dotto and Carelli, 2021). In mouse models, focal RGC axons decrease in number with mitochondrial abnormalities at the electron microscope level, but the phenotype appears slowly, not allowing for analysis in a short time frame (Alavi et al., 2007; Davies et al., 2007; Nguyen et al., 2011; Sarzi et al., 2012). Zebrafish models have displayed developmental delays in eyes and heads, short length, and body axis abnormalities (Eijkenboom et al., 2019; Rahn et al., 2013) but have not been used to investigate optic nerve degeneration.

In *C. elegans*, most studies focused on muscle cells using mutants of *eat-3*, an *OPA1* homolog (Kanazawa et al., 2008; Rolland et al., 2009). However, the mitochondria in the posterior lateral microtubule neurons did not show any difference in size from its wild-type (WT) counterparts (Byrne et al., 2019). Thus, LOF of *eat-3* may have a limited impact on phenotypic outcomes in the nervous system. Nevertheless, a DOA model in which *hOPA1*^{K301A} is expressed in GABAergic motor neurons, showed a decreased number of mitochondria in those cells (Zaninello et al., 2020). In a fly model, the *dOPA1* somatic clone in the eye exhibited a rough-eye phenotype, eye structural abnormalities, and increased MitoSOX fluorescence, which was partially rescued by vitamin E or SOD1 expression (Yarosh et al., 2008). Heterozygous *dOPA1* mutants showed shortened lifespan, elevated ROS levels, and irregular muscle tissue mitochondrial structures (Tang et al., 2009). Moreover, the Electroretinogram (ERG) pattern showed age-dependent decreases in the on transient response, heart rate, negative geotaxis response, and increased heart arrhythmia due to heat shock stress (Shahrestani et al., 2009). However, except for mice, the evidence of optic nerve axonal abnormalities is limited in model organisms such as zebrafish, worms, and flies.

The structure of the *Drosophila* visual system is similar to that of mammals (Sanes and Zipursky, 2010), with conserved mechanisms of synaptogenesis and neural circuit formation (Sanes and Zipursky, 2020). *Drosophila* photoreceptors type R7/8 extend their retinal axons from the retina directly into the brain, forming synapses in the second optic ganglion medulla via the chiasma (Figure 1A), having potential application as a model for mammalian photoreceptors and RGCs. In this study, we defined the retinal axons of *Drosophila* as analogous to the human optic nerve. Based on this, we aimed to develop an experimental model for observing and quantifying degeneration in the *Drosophila* retinal axon (Nitta et al., 2023; Richard et al., 2022).

In this study, a *dOPA1* LOF could mimic the human DOA, in which mitochondrial fragmentation, increased ROS levels, and neurodegeneration occur in retinal axons. The axonal degeneration induced by the *dOPA1* mutant was rescued by the expression of human *OPA1* (*hOPA1*). This demonstrated that the function of *hOPA1* is the same as in the fly. Previously reported mutations failed to rescue the *dOPA1* deficiency phenotype. To distinguish between the effect of LOF and DN mutations, we expressed the *hOPA1* gene of both WT and mutation in a genetic background in which *dOPA1* was deleted in retinal axons. The DOA plus mutations D438V and R445H inhibited the rescue by the WT of *hOPA1*. Taken together, we generated a new *Drosophila* DOA model which served to isolate the LOF or DN effect of *hOPA1* mutations.

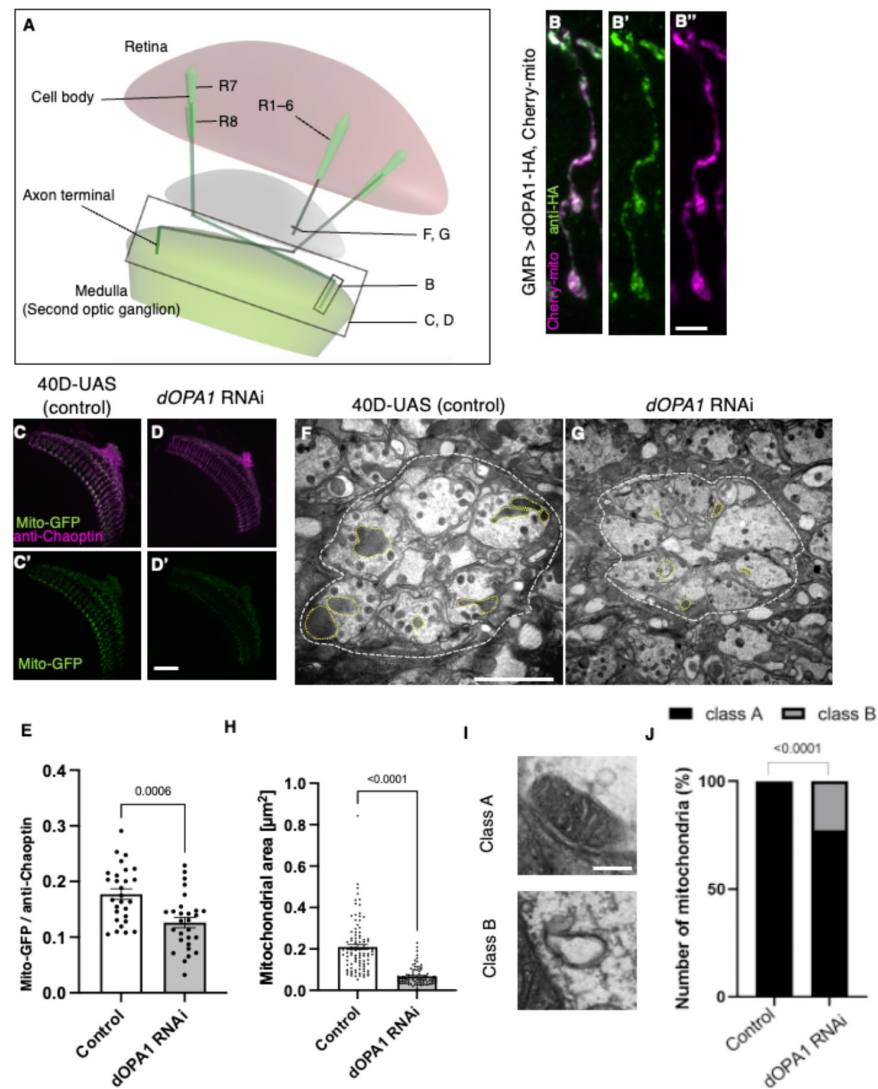


Figure 1

Effect of *dOPA1* knockdown on mitochondrial density and size in retinal axons.

(A) Schematic illustration of the *Drosophila* visual system from a dorsal view, highlighting the arrangement of photoreceptor neurons. The rectangular area indicated in B represents a single R7/8 photoreceptor axon, as illustrated in the figure for B. The rectangular areas highlighted in C and D encompass the entirety of R7/8 photoreceptor axons, as shown in the figures for C and D. The lines marked in F and G denote cross-sections of R1-6 photoreceptor axons, as depicted in the figures for F and G. (B) Visualization of mitochondria and dOPA1 in a set of R7/8 axons. Mitochondria were visualized using the expression of *mCherry-mito* driven by *GMR-Gal4* (magenta) while HA-tagged dOPA1 was immunostained using the anti-HA antibody (green). Scale bar: 3 μ m. (C, D) Representations of mitochondria visualized using the expression of *Mito-GFP* driven by *GMR-Gal4* (green), one day after eclosion. Retinal axons were labeled using the anti-Chaoptin antibody (magenta). Scale bar: 30 μ m. (E) Quantification of Mito-GFP intensity. Mito-GFP levels were calculated by dividing the Mito-GFP fluorescence intensity by the anti-Chaoptin signal intensity. Control (n = 28 optic lobes) and *dOPA1* RNAi (n = 28 optic lobes). The data are presented as mean $\pm$ SEM. (F, G) Electron micrographs of cross sections of the R1-6 retinal axons in the lamina of the Control (F) and *dOPA1* knockdown (G) on the day of eclosion. Yellow circles and white dotted lines indicate mitochondria and a lamina column, respectively. Scale bar: 2 μ m. (H) Quantification of the mitochondrial area. Control (n = 96 mitochondria) and *dOPA1* RNAi (n = 92 mitochondria). Data are expressed as mean $\pm$ SEM. (I) Representative EM images showing mitochondria with a densely packed matrix structure (classified as class A) and collapsed mitochondria (classified as class B). Scale bar: 500 nm. (J) Quantification of the mitochondria classified into class A and class B. Control (n = 96 mitochondria) and *dOPA1* RNAi (n = 92 mitochondria). See the Table1 for the genotypes of the *Drosophila* used in the experiments.

Results

dOPA1 depletion caused mitochondrial fragmentation in axon terminals

The OPA1 protein is located in the inner mitochondrial membrane. To confirm whether dOPA1 is also localized in the mitochondria of *Drosophila* retinal axons, we used a Gal4/UAS system (Brand and Perrimon, 1993 [\[1\]](#)) and expressed HA-tagged dOPA1 proteins using the eye-specific *GMR-Gal4* driver (Newsome et al., 2000a [\[2\]](#), 2000b [\[3\]](#)). The outer membrane of the mitochondria was visualized with mCherry-mito, which tagged the transmembrane domain of Miro with mCherry (Vagnoni and Bullock, 2016 [\[4\]](#)). Confocal microscopy revealed that the HA signals colocalized with mCherry-mito in the axon terminals of photoreceptor cells (Figure 1B [\[5\]](#)). This indicated that dOPA1 colocalizes with mitochondria in retinal axons.

To examine the impact of *dOPA1* on mitochondrial structure and density, we analyzed mitochondria in *dOPA1* RNAi flies. The intensity of the mitochondrial signal was significantly reduced in the retinal axons of these flies (Figure 1C and D [\[6\]](#); quantification in 1E). Previously, impaired mitochondrial transport was reported in the LOF of *marf* (MFN 1/2 homolog), which is necessary for the fusion of the outer membrane of mitochondria (Sandoval et al., 2014 [\[7\]](#)). Thus, we investigated if the LOF of *dOPA1* may cause transport defects, which may be responsible for the reduced mitochondrial density in retinal axons in *dOPA1* RNAi flies. To this end, we observed the photoreceptor neurons after puparium formation at 24 h, which allowed us to simultaneously visualize the cell bodies and axon terminals of the photoreceptors (Figure S1A [\[8\]](#)). Both *dOPA1* knockdown and control flies showed mitochondrial signals in the axon terminals (Figure S1B [\[9\]](#) and C [\[10\]](#)). However, no mitochondrial signals were observed in the axon terminals of flies with a knockdown of *milton*, which is thought to act as an adaptor between kinesin to mitochondria, promoting anterograde transport (Figure S1D [\[11\]](#)). These findings suggest that trafficking defects are not the main cause of the mitochondrial mislocalization observed in the retinal axons of the *dOPA1* knockdown.

To further analyze the mitochondrial morphology of retinal axons in the LOF of *dOPA1*, we conducted an electron microscopy analysis (Figure 1F and G [\[12\]](#)). Although *Drosophila* has eight types of photoreceptors, the axons of R1–6, which project to the lamina of the first optic lobe, can be identified without markers; thus, we chose them for this experiment. The *dOPA1* knockdown led to a significant decrease in mitochondria size in R1–6 axons (Figure 1F–H [\[13\]](#)). In addition, there was significant heterogeneity in healthy mitochondria, ranging from dense-matrix (Class A) to swollen mitochondria with a hypodense matrix (Class B) (Figure 1I [\[14\]](#)). Knocking down *dOpa1* in R1–6 axon terminals resulted in severely abnormal mitochondrial ultrastructure (22.8%; Figure 1J [\[15\]](#)). These results suggest that *dOPA1* is involved in maintaining proper mitochondrial morphology by promoting mitochondrial fusion in retinal axons.

To verify whether ROS is increased in the retinal axons in *dOPA1* knockdown photoreceptors, we measured the ROS levels in the knockdown retinal axons using MitoSOX, a mitochondrial superoxide indicator (Yarosh et al., 2008 [\[16\]](#)). The results showed significantly elevated ROS levels in *dOPA1* knockdown retinal axons compared to control (Figure 2A and B [\[17\]](#); quantification in 2C).

Given that an increase in mitophagy activity has been reported in mouse RGCs and nematode ADOA models (Zaninello et al., 2022 [\[18\]](#), 2020 [\[19\]](#)), the mitoQC marker, an established indicator of mitophagy activity, was expressed in the photoreceptors of *Drosophila*. The mito-QC reporter consists of a tandem mCherry-GFP tag that localizes to the outer membrane of mitochondria (Lee et al., 2018 [\[20\]](#)). This construct allows the measurement of mitophagy by detecting an increase in the red-only mCherry signal when the GFP is degraded after mitochondria are transported to

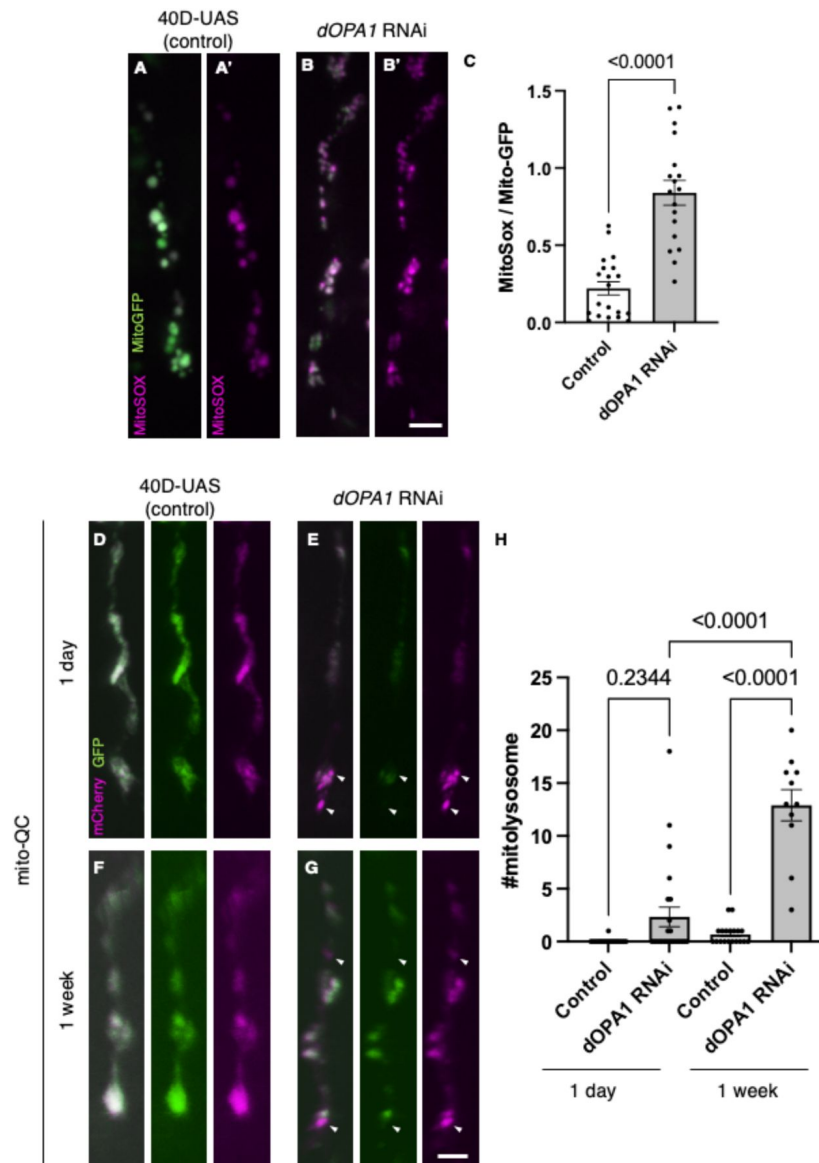


Figure 2.

elevated reactive oxygen species and mitophagy activity levels in the *dOPA1* knockdown.

(A, B) Mitochondrial reactive oxygen species (ROS) levels in a set of the R7/8 retinal axon visualized using Mito-GFP (green) driven by *GMR-Gal4* and MitoSOX (magenta) to specifically detect superoxide, an indicator of oxidative stress. Representative images of a retinal axon in the Control (A, A') and *dOPA1* knockdown (B, B'). (C) Quantification of ROS levels for each genotype. ROS levels were determined by dividing the fluorescence intensity of MitoSOX by that of Mito-GFP, highlighting the significant increase in ROS levels in *dOPA1* knockdown flies compared to controls, suggesting enhanced mitochondrial stress and dysfunction upon *dOPA1* depletion. Control (n = 20 optic lobes) and *dOPA1* RNAi (n = 18 optic lobes). Scale bar: 3 μ m. Data are presented as mean $\pm$ SEM. (D, E) Mitophagy activity in a set of the R7/8 retinal axon visualized by a genetic marker, mito-QC, driven by *GMR-Gal4*. Representative images of a retinal axon in the Control 1 day after eclosion (D), *dOPA1* knockdown 1 day after eclosion (E), the Control 1 week after eclosion (F), *dOPA1* knockdown 1 week after eclosion (G). The arrowheads indicate a mitolysosome that is positive for mCherry and negative for GFP. (H) Quantification of the mitophagy activity each genotype. The number of the mitolysosomes were counted in each genotype. The significant increase in the number of the mitolysosomes in *dOPA1* knockdown flies compared to controls after 1 week, suggesting enhanced mitophagy upon *dOPA1* depletion. Control 1d (n = 18 optic lobes), Control 1week (n = 19 optic lobe), *dOPA1* RNAi 1d (n = 24 optic lobes), and *dOPA1* RNAi 1 week (n = 11 optic lobes). Scale bar: 3 μ m. Data are presented as mean $\pm$ SEM. See the Table1 for the genotypes of the *Drosophila* used in the experiments.

lysosomes. Post *dOPA1* knockdown, we observed a significant elevation in mCherry positive and GFP negative puncta signals at one week, demonstrating an activation of mitophagy as a consequence of *dOPA1* knockdown (**Figure 2D–H** [↗](#)).

The *dOPA1* LOF leads to progressive distal degeneration of *Drosophila* photoreceptors

We tested whether knocking down *dOPA1* causes optic nerve degeneration in *Drosophila*. To this end, we performed an RNAi experiment using the *GMR-Gal4* driver and evaluated the number of retinal axons projecting to the second optic ganglion medulla (**Figure 3A** [↗](#)). To compare the number of retinal axon terminals–R7 axons–between genotypes, we used a previously developed automated method, MeDUsA (method for the quantification of degeneration using fly axons) (Nitta et al., 2023 [↗](#)). We have assessed the extent of their reduction from the total axonal terminal count, thereby determining the degree of axonal terminal degeneration (Nitta et al., 2023 [↗](#); Richard et al., 2022 [↗](#)). Our results showed that the *dOPA1* knockdown caused axonal degeneration one day after eclosion, further decreasing one week later (**Figure 3B–E** [↗](#), quantification in 3F). We obtained similar results with an independent alternative RNAi line (**Figure S2A** [↗](#) and **B** [↗](#)). To determine whether the degeneration was limited to the axons, we counted the number of rhabdomeres in the cell bodies. Each compound eye comprises 700–800 ommatidia, with each ommatidium containing eight types of photoreceptors, designated as R1–R8. The distal region of an ommatidium reveals the R1–R7 photoreceptors with a stereotypical arrangement of rhabdomeres (arrowheads in **Figure 3G–J** [↗](#)). Since it is easy to evaluate retinal degeneration by counting the number of rhabdomeres, a decrease in their number indicates degeneration of the photoreceptor cell body. The *dOPA1* knockdown caused a significant decrease in the number of rhabdomeres per ommatidium in 1-week-old adults (**Figure 3G–J** [↗](#), quantification in 3K). Interestingly, although the retinal axons had already decreased on the day of eclosion, the ommatidia remained intact (**Figure 3F and K** [↗](#)). Note that the ommatidia in R7 remained intact after one week (**Figure 3J** [↗](#)). Our results indicate that the *dOPA1* knockdown causes degeneration in both axonal terminals and cell bodies, with the neuronal degeneration starting from the axons and continuing in the cell bodies. A previous report indicated that the compound eyes of homozygous mutations of *dOPA1* displayed a glossy eye phenotype (Yarosh et al., 2008 [↗](#)). Upon knocking down *dOPA1* using the *GMR-Gal4* driver, we also observed a glossy eye-like rough eye phenotype in the compound eyes (**Figure S3** [↗](#)). The *GMR-Gal4* driver does not exclusively target Gal4 expression to photoreceptor cells. Consequently, the observed retinal axonal degeneration could potentially be secondary to abnormalities in support cells external to the photoreceptors.

To test whether our RNAi results reflect *dOPA1* downregulation, we performed mutant analysis of the *dOPA1* gene in retinal axons. For this, we used the *dOPA1*^{s3475} hypomorphic mutant allele, which has a P-element insertion in exon 2 (Yarosh et al., 2008 [↗](#)). Somatic mosaic clones were generated in the retinal axons using the FLP/FRT system (Golic and Lindquist, 1989 [↗](#)) because homozygous mutant alleles are embryonic lethal (Sandoval et al., 2014 [↗](#); Yarosh et al., 2008 [↗](#)). Note that the mutant clone analysis was conducted in a context where mutant and heterozygous cells coexist as a mosaic, and it was not possible to distinguish between them. Using MeDUsA, we found a significantly lower number of *dOPA1* mutant retinal axons than in controls one day after eclosion (**Figure 4A and B** [↗](#), quantification in 4D). To determine if *dOPA1* is responsible for axon neurodegeneration, we quantify the number of the axons in the *dOPA1* eye clone fly with the expression of *dOPA1* at one day after eclosion and found that *dOPA1* expression partially rescued the axonal degeneration (**Figure 4C** [↗](#), quantification in 4D). Our results indicate that *dOPA1* is necessary for maintaining the number of retinal axons in the *Drosophila* visual system. In conclusion, the neurodegeneration observed in the LOF of *dOPA1* is due to the progressive loss of retinal axons, as well as the mammal optic nerve.

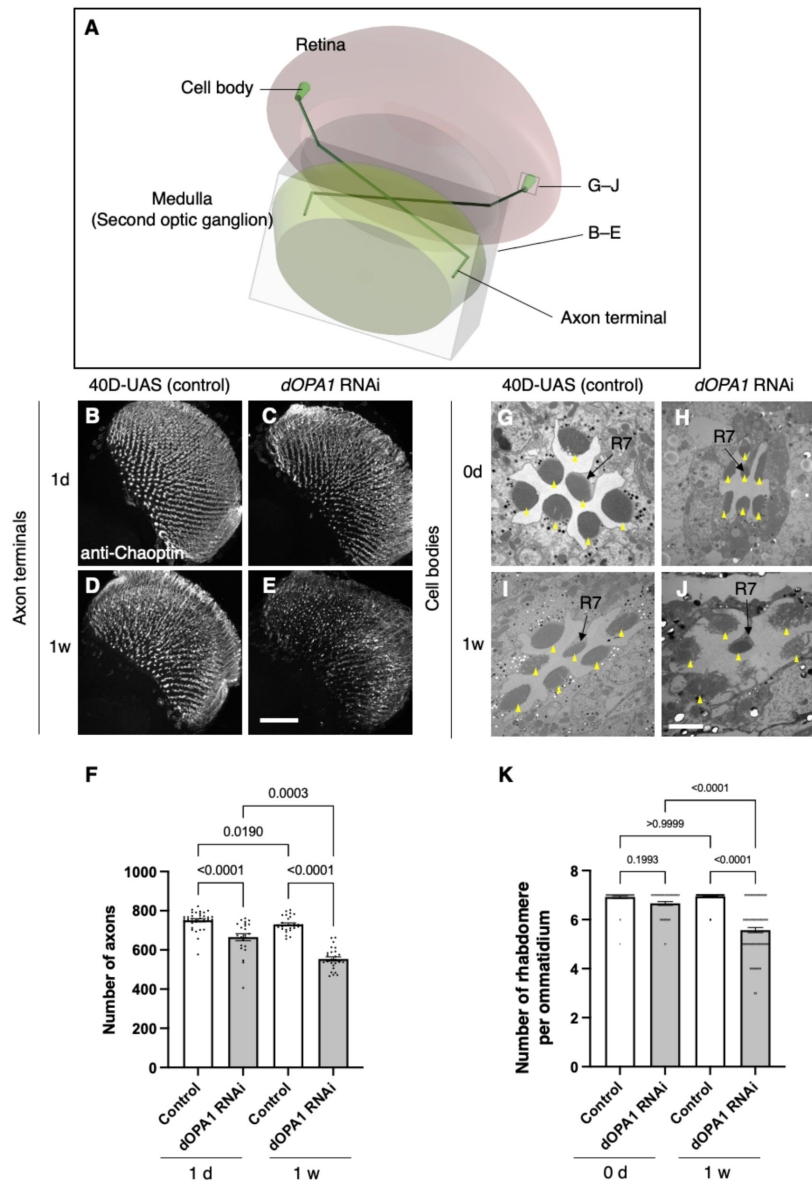


Figure 3.

Effect of *dOPA1* knockdown on photoreceptor neurodegeneration in *Drosophila*.

Schematic of the visual system in *Drosophila*. A set of R7/8 photoreceptors project their axons to the second optic ganglion medulla. The rectangular prism indicated in B–E represents the area of the whole medulla, as shown in the figure for B–E. The square marked in G–J denote cross-sections of the cell bodies of the photoreceptors, as depicted in the figures for G–J. (B–E) All R7 and R8 axon terminals project to the medulla. *dOPA1* in the photoreceptor was knocked down by *GMR-Gal4*. *40D-UAS* was used as a Control to match the number of *UAS* sequences recognized by *Gal4*. One day after eclosion in Control (B) and *dOPA1* knockdown (C), and one week after in Control (D) and *dOPA1* knockdown (E). The retinal axons were stained with anti-Chaoptin, a photoreceptor-specific antibody. Scale bar = 50 μ m. (F) Number of axons at each time point in R7 neurons for each situation quantified by MeDUSA. At 1 day of Control ($n = 34$ optic lobes) and *dOPA1* RNAi ($n = 23$ optic lobes), and at 1 week of Control ($n = 28$ optic lobes) and *dOPA1* RNAi ($n = 26$ optic lobes). Data are expressed as mean $\pm$ SEM. (G–J) Electron micrographs of cross sections of the photoreceptor cell bodies in the ommatidia of the retina. Rhabdomeres are shown as yellow arrowheads. The day of eclosion in Control (G) and *dOPA1* knockdown (H), and 1-week-old adults of Control (I) and *dOPA1* knockdown (J). Scale bar = 2 μ m. (K) Quantification of the number of rhabdomeres for each genotype and time point. At 0 day in Control ($n = 50$ ommatidia) and *dOPA1* RNAi ($n = 127$ ommatidia), and at 1 week in Control ($n = 50$ ommatidia) and *dOPA1* RNAi ($n = 117$ ommatidia). Data are expressed as mean $\pm$ SEM. See the Table1 for the genotypes of the *Drosophila* used in the experiments.

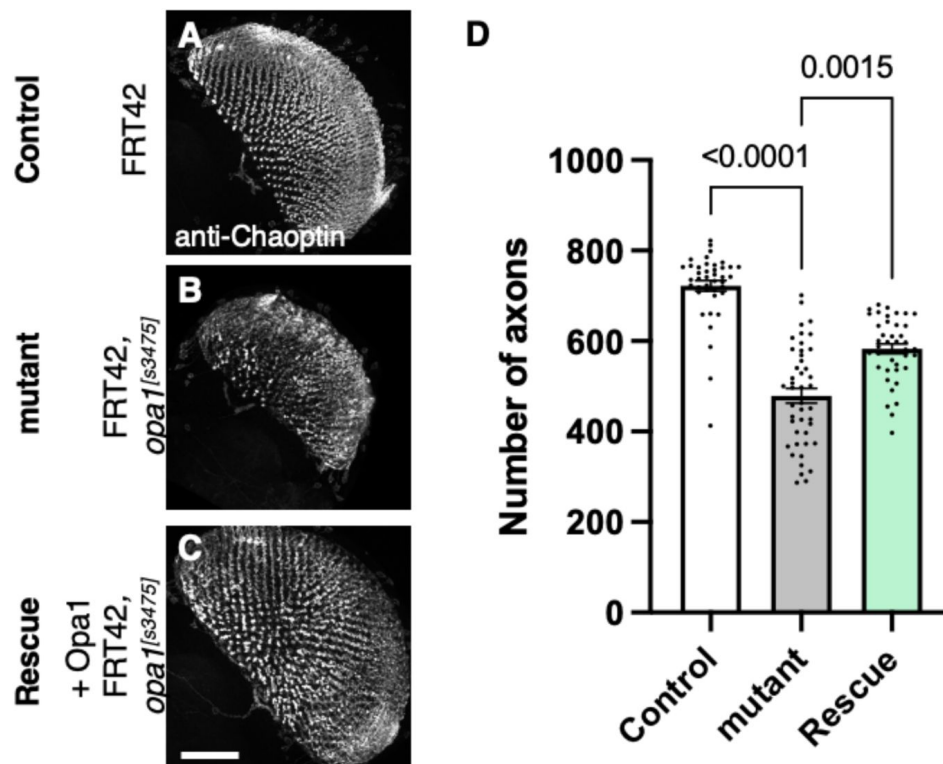


Figure 4.

Role of *dOPA1* in the retinal axon according to mutant analysis.

(A–C) Representations of retinal axons labeled with an anti-Chaoptin antibody (gray), one day after eclosion. The images show Control (A), *dOPA1*^{s3475} somatic mosaic flies (B), and *dOPA1*^{s3475} somatic mosaic flies expressing the eye-specific full-length *dOPA1* (C). Scale bar: 50 μ m. (D) Quantification of the number of axons for each genotype. Control (n = 41 optic lobes), mutant (n = 45 optic lobes), and rescue (n = 41 optic lobes). The data are presented as mean $\pm$ SEM. See the Table1 for the genotypes of the *Drosophila* used in the experiments.

Investigating disease mutations in OPA1 using a fly DOA model to confirm their pathological significance

Our observations in *dOPA1* LOF flies confirmed mitochondria fragmentation, increased ROS levels, and degeneration of retinal axons, as reported in mammals. These data indicated that the function of *OPA1* may be conserved across *Drosophila* and humans. Thus, we tested whether *hOPA1* could effectively replace *dOPA1* in flies by generating a transgenic organism to express *hOPA1* in *Drosophila*. Following the UAS sequence, that the yeast-derived transcription factor Gal4 binds, the construct included an HA tag, followed by the *hOPA1* gene, and a myc tag (**Figure 5A** [↗](#)). This construct was expressed in *Drosophila* and protein expression was confirmed by Western blotting. The HA tag, attached to the N-terminus, was immunoblotted resulting in a band of the expected full-length size (**Figure 5B** [↗](#)). For the myc tag at the C-terminus, a strong band and two weak bands were observed at the upper and lower positions (**Figure 5B** [↗](#)). Comparison of the expression levels across the variants revealed no significant differences in protein expression (**Figure 5C** [↗](#)). We also employed a human anti-OPA1 antibody to verify the expression of *hOPA1* in vivo in *Drosophila*, detecting both the long and short forms of OPA1, band 2 and band 4 respectively (**Figure 5D** [↗](#)).

Using this *UAS-hOPA1*, we performed rescue experiments. *hOPA1* expression in the retinal axons of the *dOPA1* mutant, which generated somatic cell clones, partially rescued the number of axons (**Figure 5E** [↗](#)). These results suggest that *hOPA1* and *dOPA1* are interchangeable. We also generated and expressed the *hOPA1* mutation 2708-2711del, which is known to cause DOA, the I382M mutation, located in the GTPase domain and associated with DOA, as well as the D438V and R445H mutations, also located in the GTPase domain and associated with DOA plus. The expression of these mutations in retinal axons failed to rescue the *dOPA1* deficiency to the same extent as the WT *hOPA1* (**Figure 5E** [↗](#)). Importantly, unlike the D438V and R445H mutations, the 2708-2711del and I382M mutations could be weakly rescued. This suggests that these mutations result in LOF, but may retain some residual activity. These findings are also in line with the observed severity of the associated diseases, DOA and DOA plus. As the D438V and R445H mutations were not significantly rescued, unlike the 2708-2711del and I382M mutations. Our methodology distinctively facilitates the quantitative evaluation of LOF severity by comparing the rescue capabilities of various mutations. Notably, the 2708-2711del and I382M mutations demonstrated only partial rescue, indicative of a hypomorphic effect with residual activity. In contrast, the D438V and R445H mutations failed to show significant rescue, suggesting a more profound LOF. The correlation between the partial rescue by the 2708-2711del and I382M mutations and their classification as hypomorphic is significant. Moreover, the observed differences in rescue efficacy correspond to the clinical severities associated with these mutations, namely in DOA and DOA plus disorders. Thus, our results substantiate the model's ability to quantitatively discriminate among mutations based on their impact on protein functionality, providing an insightful measure of LOF magnitude.

Furthermore, we assessed the potential for rescuing ROS signals. Similar to its effect on axonal degeneration, wild-type *hOPA1* effectively mitigated the phenotype, whereas the 2708-2711del, D438V, and R445H mutants did not (**Figure 5F** [↗](#)). Importantly, the I382M variant also reduced ROS levels comparably to the wild type. These findings demonstrate that both axonal degeneration and the increase in ROS caused by *dOPA1* downregulation can be effectively counteracted by *hOPA1*. Although I382M retains partial functionality, it acts as a relatively weak hypomorph in this experimental setup.

We also conducted Western blot analyses using anti-COXII and anti-Atg8a antibodies to assess changes in mitochondrial quantity and autophagy activity following the knockdown of *dOPA1*. Mitochondrial protein levels, indicated by COXII quantification, were evaluated to verify

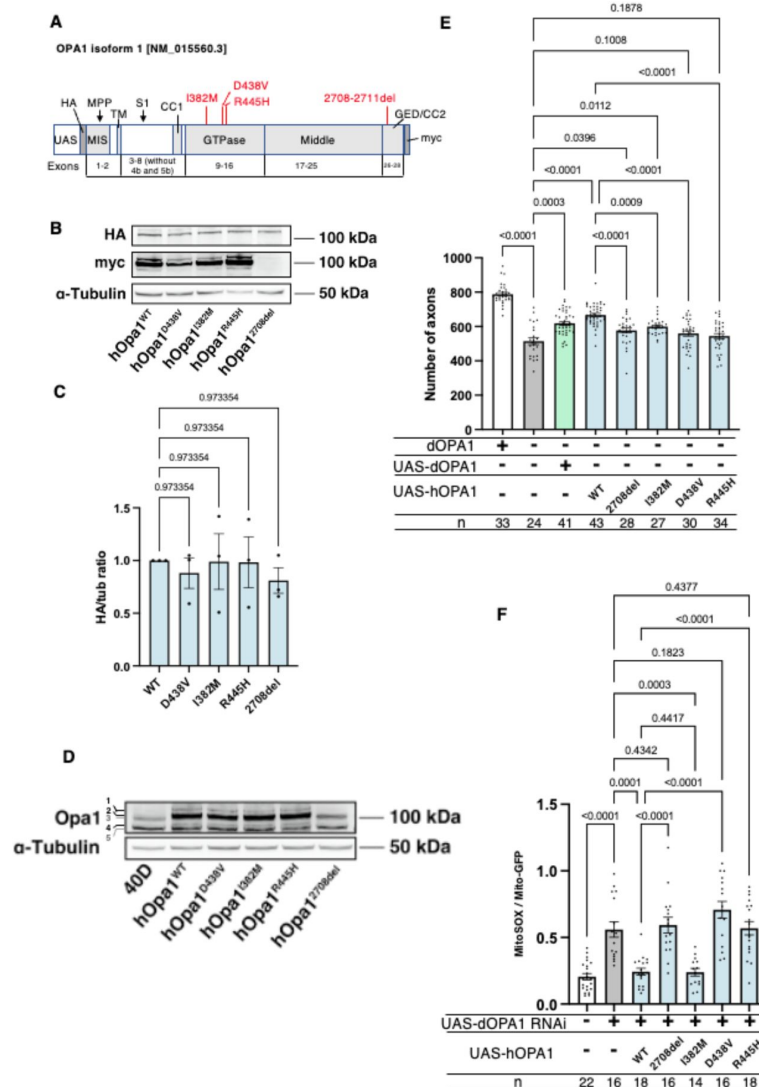


Figure 5.

Verification of the pathological significance of disease mutations in *hOPA1*.

(A) Schematic illustration of the *hOPA1* gene construct with HA and myc tags in the UAS-based vector. *hOPA1* includes a mitochondrial import sequence (MIS) cleaved by mitochondrial processing peptidase (MPP), a transmembrane region (TM), a coiled-coil region (CC1), a GTPase domain, a Middle domain, and a GTPase effector domain (GED) containing a coiled-coil region (CC2). The sites of variants (I382M, D438V, R445H, and a deletion from 2708 to 2711) are shown in red. S1 is the site cleaved by OMA1. (B) Western blot analysis to confirm the expression of *hOPA1* variants. *hOPA1*_WT, *hOPA1*_D438V, *hOPA1*_I382M, *hOPA1*_R445H, and *hOPA1*_2708del were expressed in whole *Drosophila* bodies and detected using anti-HA and anti-myc antibodies. α -Tubulin was used as a loading Control. (C) Quantification of the expression levels of the OPA1 protein for each variant. *hOPA1*_WT (n = 3), D438V (n = 3), I382M (n = 3), R445H (n = 3), and 2708del (n = 3). The data are presented as mean $\pm$ SEM. (D) Detection of *hOPA1* expression in vivo in *Drosophila* using anti-OPA1. Band 1 represents the full-length OPA1, which includes the MIS, while Band 2 corresponds to the long-form OPA1 (L-OPA1), cleaved at the MPP. Bands 3 and 5 potentially detect the endogenous *dOPA1*. Band 4 is thought to represent the short-form OPA1 (S-OPA1). (E) Rescue experiments were conducted to assess the expression of each *hOPA1* variant, including *dOPA1*, in the retina axons of *dOPA1* mutant somatic clones. The sample size is indicated (n). Data are presented as mean $\pm$ SEM. (F) Quantification of mitochondrial ROS levels upon expression of *hOPA1* variants in the context of *dOPA1* knockdown. Control (n = 22 optic lobes), *dOPA1* RNAi (n = 16 optic lobes), *hOPA1*_WT with *dOPA1* RNAi (n = 18 optic lobes), *hOPA1*_2708-2711del with *dOPA1* RNAi (n = 16 optic lobes), *hOPA1*_I382M with *dOPA1* RNAi (n = 14 optic lobes), *hOPA1*_D438V with *dOPA1* RNAi (n = 16 optic lobes), and *hOPA1*_R445H with *dOPA1* RNAi (n = 18 optic lobes). Data are presented as mean $\pm$ SEM. See the Table1 for the genotypes of the *Drosophila* used in the experiments.

mitochondrial content, and the ratio of Atg8a-1 to Atg8a-2 was used to measure autophagy activation. For these experiments, *Tub-Gal4* was employed to systemically knockdown *dOPA1*. Considering the lethality of a whole-body *dOPA1* knockdown, *Tub-Gal80^{TS}* was utilized to repress the knockdown until eclosion by maintaining the flies at 20°C. After eclosion, we increased the temperature to 29°C for two weeks to induce the knockdown or expression of *hOPA1* variants. The results revealed no significant differences across the genotypes tested (**Figure S4A** [↗](#)–**D** [↗](#)).

Distinction between LOF and DN mutations in *hOPA1* linked to DOA or DOA plus

Currently, the *hOPA1* mutations that contribute to DOA plus have been primarily located in the GTPase domain of *hOPA1*. The role of *hOPA1* as a GTPase is facilitated by its ability to form a polymer via the GED domain and Middle domain (Li et al., 2019 [↗](#)). This led to the hypothesis that mutations in the GTPase domain can interact with WT *hOPA1* but cannot activate the GTPase activity to show a DN effect. However, it has been challenging to clarify whether these mutations are DN or LOF, as there is a significant number of LOF mutations in the GTPase domain. Given that *dOPA1* can be substituted with *hOPA1*, we suspected that a DN effect could also be achieved by expressing the D438V or R445H mutations in the WT background. However, the axons did not degenerate despite expression and monitoring for two weeks in adult flies (**Fig. 6A** [↗](#)). The results presented in **Figure 5C** [↗](#) indicate that there are no significant differences in the expression levels among the variants, suggesting that variations in expression levels do not influence the outcomes. The amino acid sequences of *hOPA1* and *dOPA1* are highly conserved (72% identity), particularly in the GTPase domain. The Middle and GED domains are also relatively well-preserved, with 54% and 64% agreement, respectively (**Figure S5** [↗](#)). These results imply that, although *dOPA1* and *hOPA1* are functionally interchangeable, they may not interact with each other. Consequently, we expressed *hOPA1* in the photoreceptor depleted of *dOPA1* by somatic clone; under these conditions, we further expressed *hOPA1* WT, 2708-2711del, and mutations in the GTPase domains, including R445H, I382M, and D438V. This allowed *hOPA1* to interact with itself and to verify the DN effect. As a result, the expression of the WT, 2708del, and I382M did not result in any changes when rescued by *hOPA1* substitution (**Figure 6B** [↗](#)). However, the rescue was significantly suppressed for D438V and R445H, which are known to cause DN; thus, the DN effect could be replicated (**Figure 6B** [↗](#)). In conclusion, we established an experimental model that can separate LOF and DN, the pathological significance of *hOPA1* mutations. We investigated the impacts of dominant negative mutations on mitochondrial oxidation levels, mitochondrial quantity, and autophagy activation levels; however, none of these parameters showed statistical significance (**Figure S6** [↗](#)).

Discussion

In this study, we found that LOF of *dOPA1* in *Drosophila* can imitate human DOA, in which the optic nerve degenerates. We were successful in reversing this degeneration by expressing *hOPA1* (**Figure 6C** [↗](#)). However, we could not rescue any previously reported mutations known to cause either DOA or DOA plus except for I382M. In the context of rescuing the retinal axons of the *dOPA1* mutant by expressing the WT of *hOPA1*, it was observed that only the DOA plus mutations suppressed the rescue (**Figure 6C** [↗](#)). This allowed us to distinguish between LOF and DN mutations in *hOPA1*. The fly model developed in this study will allow investigating new *hOPA1* mutations for DOA or DOA plus. We have previously utilized MeDUSA to quantify axonal degeneration, applying this methodology extensively to various neurological disorders. The robust adaptability of this experimental system is demonstrated by its application in exploring a wide spectrum of genetic mutations associated with neurological conditions, highlighting its broad utility in neurogenetic research. We identified a novel de novo variant in Spliceosome Associated Factor 1, Recruiter of U4/U6.U5 Tri-SnRNP (*SART1*). The patient, born at 37 weeks with a birth weight of 2934g, exhibited significant developmental delays, including an inability to support head

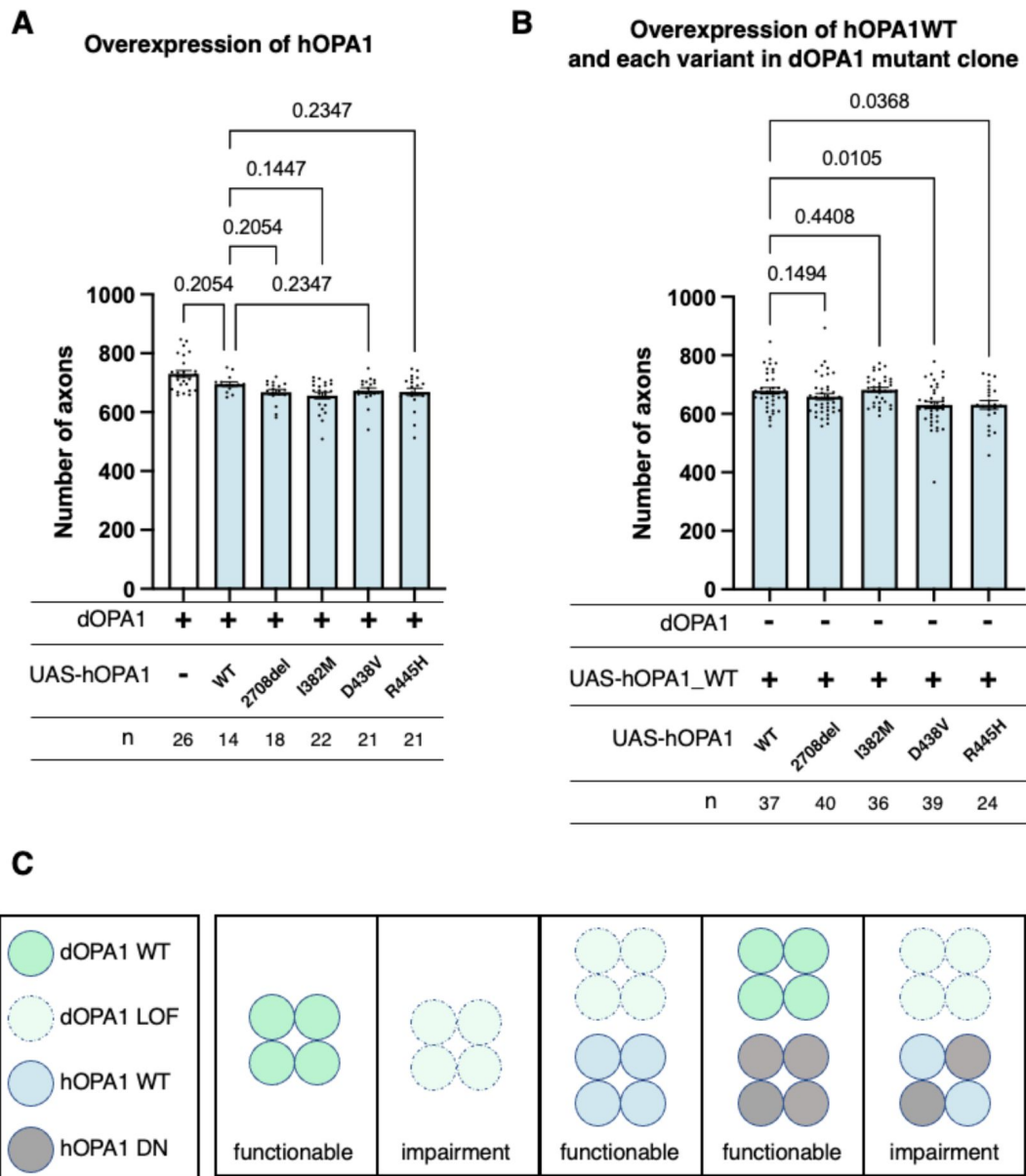


Figure 6.

LOF or DN effects of disease mutations in *hOPA1*.

(A) Impact of each human OPA1 variant on the axon number in the optic nerve of *Drosophila* as quantified using MeDUSA. (B) Expression of both *hOPA1* WT and its respective variants was analyzed in photoreceptors lacking *dOPA1*, and the number of axons was quantified using the MeDUSA. The sample size is indicated (n). Data are expressed as mean $\pm$ SEM. (C) Schematic representation of interspecies differences in OPA1 interactions and the interchangeability of OPA1 between human and fly. See the Table1 for the genotypes of the *Drosophila* used in the experiments.

movement at 7 months, reliance on tube feeding, unresponsiveness to visual stimuli, and development of infantile spasms with hypsarrhythmia, as evidenced by EEG findings. Profound hearing loss and brain atrophy were confirmed through MRI imaging. To assess the functional impact of this novel human gene variant, we engineered transgenic *Drosophila* lines expressing both wild type and mutant *SART1* under the control of a UAS promoter. Our MeDUSA analysis suggested that the variant may confer a gain-of-toxic-function (Nitta et al., 2023 [↗](#)). Moreover, we identified heterozygous loss-of-function mutations in *DHX9* as potentially causative for a newly characterized neurodevelopmental disorder. We further investigated the pathogenic potential of a novel heterozygous de novo missense mutation in *DHX9* in a patient presenting with short stature, intellectual disability, and myocardial compaction. Our findings indicated a loss of function in the G414R and R1052Q variants of *DHX9* (Yamada et al., 2023 [↗](#)). This experimental framework has been instrumental in elucidating the impact of gene mutations, enhancing our ability to diagnose how novel variants influence gene function. Our research established that *dOPA1* knockdown precipitates axonal degeneration and elevates ROS signals in retinal axons. Expression of human *OPA1* within this context effectively mitigated both phenomena; it partially reversed axonal degeneration and nearly completely normalized ROS levels. These results imply that factors other than increased ROS may drive the axonal degeneration observed post-knockdown. Furthermore, while differences between the impacts of DN mutations and loss-of-function mutations were evident in axonal degeneration, they were less apparent when using ROS as a biomarker. The extensive use of transgenes in our experiments might have mitigated the knockdown effects. In a systemic *dOPA1* knockdown, assessments of mitochondrial quantity and autophagy activity revealed no significant changes, suggesting that the cellular consequences of reduced *OPA1* expression might vary across different cell types.

In our study, expressing *dOPA1* in the retinal axons of *dOPA1* mutants resulted in significant rescue, but it did not return to control levels. There are three possible explanations for this result. The first concerns gene expression levels. The Gal4-line used for the rescue experiments may not replicate the expression levels or timing of endogenous *dOPA1*. Considering that the optimal functionality of *dOPA1* may be contingent upon specific gene expression levels, attaining a wild-type-like state necessitates the precise regulation of these expression levels. The second is a non-autonomous issue. Although *dOPA1* gene expression was induced in the retinal axons for the rescue experiments, many retinal axons were homozygous mutants, while other cell types were heterozygous for the *dOPA1* mutation. If there is a non-autonomous effect of *dOPA1* in cells other than retinal axons, it might not be possible to restore the wild-type-like state fully. The third potential issue is that only one isoform of *dOPA1* was expressed. In mouse *OPA1*, to completely restore mitochondrial network shape, an appropriate balance of at least two different isoforms, l-*OPA1* and s-*OPA1*, is required (Del Dotto et al., 2017 [↗](#)). This requirement implies that multiple isoforms of *dOPA1* are essential for the dynamic activities of mitochondria.

Our established fly model is the first simple organism to allow observation of degeneration of the retinal axons. The mitochondria in the axons showed fragmentation of mitochondria. Former studies have observed mitochondrial fragmentation in S2 cells (McQuibban et al., 2006 [↗](#)), muscle tissue (Deng et al., 2008 [↗](#)), segmental nerves (Trevisan et al., 2018 [↗](#)), and ommatidia (Yarosh et al., 2008 [↗](#)) due to the LOF of *dOPA1*. Our study presents compelling evidence that *dOPA1* knockdown instigates neuronal degeneration, characterized by a sequential deterioration at the axonal terminals and extending to the cell bodies. This degenerative pattern, commencing from the distal axons and progressing proximally towards the cell soma, aligns with the paradigm of 'dying-back' neuropathy, a phenomenon extensively documented in various neurodegenerative disorders (Wang et al., 2012 [↗](#)). Previously, the function of *OPA1* in mice (*mOPA1*) was substituted by *hOPA1* (Del Dotto et al., 2017 [↗](#); Sarzi et al., 2018 [↗](#)). Here we demonstrated that the function of *dOPA1* can also be substituted by *hOPA1*. Thus, the function of *OPA1* is conserved across a wide range of species but our fly model has the advantage of allowing rapid observation of phenotypes. In the mouse model of DOA with the heterozygous deletion of the *mOPA1* gene at positions 329–355 (Alavi et al., 2007 [↗](#)) or delTTAG mutant (Sarzi et al., 2012 [↗](#)), a decrease in the number of RGC

was observed after 17 months or 16 months of age, respectively. To save time, using a fly model for quick analysis and direct observation of retinal axons would be useful. Previous models using nematodes, flies, and zebrafish did not observe axonal degeneration directly, but our new model has made this possible. As a result, our model has the potential to be used for screening for modifiers of unclear molecular pathologies and drug screening.

Our model could also demonstrate the pathological significance of *hOPA1* mutations and be used for genetic diagnosis to differentiate between LOF and DN mutations. To elucidate the pathophysiological implications of mutations in the *OPA1* gene, we engineered and expressed several human *OPA1* variants, including the 2708-2711del mutation, associated with DOA, and the I382M mutation, located in the GTPase domain and linked to DOA. We also investigated the D438V and R445H mutations in the GTPase domain and correlated with the more severe DOA plus phenotype. The 2708-2711del mutation exhibited limited detectability via HA-tag probing. Still, it was undetectable with a myc tag, likely due to a frameshift event leading to the mutation's characteristic truncated protein product, as delineated in prior studies (Zanna et al., 2008). Contrastingly, the I382M, D438V, and R445H mutations demonstrated expression levels comparable to the WT *hOPA1*.

However, the expression of these mutants in retinal axons did not restore the *dOPA1* deficiency to the same extent as the WT *hOPA1*, as evidenced in **Figure 5E**. This finding indicates a functional impairment imparted by these mutations, aligning with established understanding (Zanna et al., 2008). Notably, while the 2708-2711del and I382M mutations exhibited limited functional rescue, the D438V and R445H mutations did not show significant rescue activity. This differential rescue efficiency suggests that the former mutations, particularly the I382M, categorized as a hypomorph (Del Dotto et al., 2018), may retain partial functional capacity, indicative of a LOF effect but with residual activity. The I382M missense mutation within the GTPase domain of *OPA1* has been described as a mild hypomorph or a disease modifier. Intriguingly, this mutation alone does not induce significant clinical outcomes, as evidenced by multiple studies (Bonifert et al., 2014; Bonneau et al., 2014; Carelli et al., 2015; Schaaf et al., 2011). A significant reduction in protein levels has been observed in fibroblasts originating from patients harboring the I382M mutation. However, mitochondrial volume remains unaffected, and the fusion activity of mitochondria is only minimally influenced (Del Dotto et al., 2018; Kane et al., 2017). This observation is consistent with findings reported by de la Barca et al. (Chao De La Barca et al., 2020), where a targeted metabolomics approach classified I382M as a mild hypomorph. In our current study, the I382M mutation preserves more *OPA1* function compared to DN mutations, as depicted in **Figures 5E and F**. Considering the results from our *Drosophila* model and previous research, we hypothesize that the I382M mutation may constitute a mild hypomorphic variant. This might explain its failure to manifest a phenotype on its own, yet its contribution to increased severity when it occurs in compound heterozygosity.

Our results for LOF or DN segmentation were consistent with previously reported yeast models (Del Dotto et al., 2018) and fibroblasts obtained from patients with *OPA1* mutations (Kane et al., 2017), and confirmed the conserved function of *OPA1* across species. A major advantage of the *Drosophila* model over yeast or fibroblasts, is that we can observe a phenotype that mimics the actual disease state of axonal degeneration in retinal axons and analyze it not only at the molecular level, but also the structural level of axons and mitochondria. The transgenic *Drosophila* of disease-associated *hOpa1* mutations created in this study can also serve to understand the pathophysiology of these mutations.

Advances in genome analysis allowed the identification of many variants and genetic mutations associated with DOA using next-generation sequencing. This could confirm diagnoses and identify new forms of DOA. The widespread use of these sequencers has also created new challenges, such as the growing number of variants of unknown significance (VUS) in publicly available databases such as ClinVar for DOA. These variants are categorized as pathogenic, likely pathogenic, VUS,

likely benign, or benign, and the number of VUS is increasing as more variants are registered. The ClinVar database had classified 110 out of 319 *Opa1* variants as VUS at the end of 2019. By the end of 2022, this number had increased to 357 out of 840, indicating an increase from 34% to 42% over the last three years (Henrie et al., 2018). This trend suggests that the number of VUS in *Opa1* may continue to increase in the future. Analyzing VUS requires testing the effects of a mutation *in vitro* or *in vivo*; this can be expensive and time-consuming without prior confirmation or high probability of pathological significance. However, our fly model can be used to easily and inexpensively determine the pathological effects of disease-causing gene mutations among the growing number of rare mutations identified in DOA.

In this study, we tested the effect of expressing the D438V and R445H mutations in the *dOPA1* background and found that retinal axons did not degenerate. Regarding the interactions among OPA1 proteins, in yeast Mgm1, K854 in the region near the C-terminus is important for aggregation, and it becomes an oligomer to increase the activity of GTPase (Rujiviphat et al., 2009). Although hOPA1 normally has low GTPase activity, its interaction with negatively charged phospholipids such as cardiolipin causes hOPA1 to aggregate, increasing GTP hydrolysis activity (Ban et al., 2010). In the thermophilic fungus *Chaetomium thermophilum*, Mgm1 forms a dimer and then a tetramer at the C-terminal Stalk domain of Mgm1 (Faelber et al., 2019). Structural analysis also predicted that the Middle and GED domains are necessary for dimer and oligomer formation in hOPA1 (Li et al., 2019). In addition, peptide-binding assays have revealed that the coiled-coil domain, which is part of the GED domain, is required for hOPA1 interactions (Akepati et al., 2008). Comparing the amino acid sequences of hOPA1 and dOPA1, the Middle and GED domains, which are necessary for interactions between OPA1 molecules, exhibit concordance rates of 54.3% and 63.6%, respectively. These percentages are low compared to the 71.8% identity observed in the GTPase domains. These suggest that the OPA1 homologs of each species form oligomers with their unique OPA1. Note that the expression of *dOPA1*^{K273A}, a presumably GTPase-negative form of *dOPA1* (Tsuyama et al., 2017), degenerated retinal axons (Figure S7). These results imply that, while dOPA1 and hOPA1 are interchangeable, they may not interact with each other. This could pave the way for the development of a chimeric hOPA1, which would retain its functional properties while avoiding the interaction with endogenous hOPA1. Such chimeric hOPA1 could potentially be used as gene therapy since induced pluripotent stem cells have been generated from a patient with R445H (Sladen et al., 2021).

OPA1 has been implicated in various other diseases as well, including normal tension glaucoma (Powell et al., 2003; Yu-Wai-Man et al., 2010b), multiple sclerosis-like illness (Verny et al., 2008; Yu-Wai-Man et al., 2016), Parkinson's disease and dementia (Carelli et al., 2015; Lynch et al., 2017), and cardiomyopathy (Spiegel et al., 2016). Although the connection between these diseases and OPA1 is not fully understood, further research on DOA may provide a deeper insight into this relationship. In the future, pathological analysis using the present model could have implications for understanding the mechanisms underlying these diseases.

DOA has also been reported to involve mitochondrial DNA (mtDNA) depletion (Amati-Bonneau et al., 2008), but our model, while useful for nuclear gene analysis, is limited for mtDNA analysis. The gene content in the *Drosophila* mtDNA genome is similar as that in vertebrates, but the gene order and distribution on both DNA strands differ. mOPA1 interacts with mtDNA nucleoids through exon 4b binding to mtDNA D-loops, independent of mitochondrial fusion (Yang et al., 2020). However, the homologous region of mOPA1 or hOPA1 to exon4b is not found in *Drosophila*, and the D-loop of mtDNA has not been found in cultured cells of flies or in mtDNA of embryos using electron microscopy (Rubenstein et al., 1977). In addition, while the D-loop is involved in mtDNA replication in humans (Fish et al., 2004), the A + T region is the origin of replication in mtDNA replication in *Drosophila* (Goddard and Wolstenholme, 1978). Thus, there may be differences in the mechanism of mtDNA replication between humans and flies (Garesse

and Kaguni, 2005 [\[10\]](#)). Therefore, the association of dOPA1 with mtDNA may differ from that in mammals and the regulatory mechanism of mtDNA homeostasis in which dOPA1 is involved requires further investigation.

Another limitation of our model is that the molecular mechanisms underlying the formation of long-form OPA1 (L-OPA1) and short-form OPA1 (S-OPA1), which are well analyzed in mammals (MacVicar and Langer, 2016 [\[11\]](#)), have not been elucidated in *Drosophila*. In hOPA1, S1 or S2 sites are processed by the i-AAA protease OMA1 and Yme1L respectively to produce S-hOPA1 (Anand et al., 2014 [\[12\]](#)). Despite the fact that Yme1L is conserved in *Drosophila*, it has been involved in the degradation of the dOPA1 protein rather than its cleavage by dYme1L (Liu et al., 2020 [\[13\]](#)). Moreover, the cleavage site for i-AAA does not exist in dOPA1 (Olichon et al., 2007 [\[14\]](#)). Regarding other proteases, yeast Mgm1 is processed by the rhomboid protease Pcp1 (Esser et al., 2002 [\[15\]](#); Herlan et al., 2003 [\[16\]](#)). However, the Rhomboid-7, a Pcp1 ortholog of *Drosophila*, was not required for dOPA1 processing (Whitworth et al., 2008 [\[17\]](#)). Although hOPA1 expression in *Drosophila* has identified three different sizes of hOPA1 in our Western blotting result, it is unclear how they are cleaved. From a size perspective, the upper band was inferred to represent the full-length hOPA1 including the mitochondria import sequence (MIS). Since the Middle and the lowest bands were not present in HA-tagged samples, it is possible that the bands at the Middle size represent the long-form hOPA1 (L-hOPA1) in which probably MIS was processed. The band detected at the lowest position may represent a short form of hOPA1 (S-hOPA1). However, *Drosophila* does not have OMA1, an i-AAA protease that cleaves the S1 site. Although another i-AAA protease that cleaves the S2 site, YME1L, is conserved, our expressed hOPA1 is isoform 1 and lacks the S2 site (Fig. 5A [\[18\]](#)). Therefore, it is not clear how S-hOPA1 is processed in *Drosophila*. The anti-dOPA1 antiserum detects three bands in the overexpression of a FLAG-tagged dOPA1 construct (Poole et al., 2010 [\[19\]](#)), suggesting a molecular mechanism by which dOPA1, like yeast and mammals, is cleaved after translation. However, the proteases involved in the mechanism are not yet clarified. Whether *Drosophila*'s endogenous mechanisms can be used to study how L-OPA1 and S-OPA1 are involved in DOA remains unclear.

In this study, we established a new model of DOA in *Drosophila*, in which we discovered the following: 1. We could replicate the human disease of optic nerve degeneration, allowing rapid genetic disease analysis. 2. We could distinguish the LOF and DN effects of *hOPA1* by observation of axonal degeneration phenotype but not with ROS signal, autophagy activity, and amount of the mitochondria in this model. These findings can reveal the pathological significance of *de novo* mutations and are useful in diagnosing DOA or DOA plus. Additionally, while *hOPA1* is the major gene involved in DOA, other genes involved in DOA and interacting proteins have not been investigated yet. Our model can be used for pathological modifier screening, exploring modifiers, and contributing to drug development. In the future, we hope to provide insights that can be applied to the fundamental treatment of DOA.

Materials and Methods

Fly strains

Flies were maintained at 25°C on standard fly food. For knockdown experiments (Figures 1C–E [\[20\]](#), 1F–J, 2A–H, 3B–K, 5F, S1, S2, S6A, and S7), flies were kept at 29°C in darkness. Female flies were used in all experiments to ensure consistency in the number of retinal axons. The following fly strains were obtained from the Bloomington Drosophila Stock Center (BDSC): *GMR-Gal4* (BDSC 1104), *Tub-Gal4* (BDSC 5138), *UAS-dicer2(III)* (BDSC 24651), *UAS-mito-HA-GFP* (BDSC 8443), *UAS-mCherry-mito* (BDSC 66532), *UAS-mitoQC* (BDSC 91641), *UAS-dOPA1 RNAi* (BDSC 32358), *UAS-Milton RNAi* (BDSC 44477), *FRT42D*, *GMR-hid*, *l(2)cl-R11[1]/CyO*; *ey-Gal4*, *UAS-flp* (BDSC 5251), *ey3.5-flp* (BDSC 35542), *Tub-Gal80^{TS}* (BDSC 7017), and *UAS-luciferase RNAi* (BDSC 31603).

The following fly strains were obtained from the Vienna Drosophila Resource Center (VDRC): The *40D-UAS* (VDRC 60101) and *UAS-dOPA1 RNAi* (VDRC 106290). The following fly strains were obtained from the Kyoto Drosophila Stock Center (Kyoto): *FRT19* (Kyoto 101231), *FRT42D* (Kyoto 101878), and *dOPA1^{s3475}* (Kyoto 111438). The *UAS-dOPA1-HA* and *UAS-dOPA1-HA^{K273A}* were generously provided by T. Uemura (Kyoto University, Japan). The *dOPA1* clone analysis was performed by inducing flippase expression in the eyes using either *ey-Gal4* with *UAS-flp* or *ey3.5-flp*, followed by recombination at the chromosomal location *FRT42D* to generate a mosaic of cells homozygous for *dOPA1^{s3475}*.

Generation of transgenic flies

The cDNA for WT, I382M, D438V, R445H, and 2708-2711del forms of *hOPA1* [NM_015560.3] were generated by Vectorbuilder Inc. (Yokohama, Japan). Then, the HA-hOPA1-myc sequence was amplified and inserted into pJFRC81-10XUAS-IVS-Syn21-GFP-p10 (ID36432, Addgene, USA) digested with *NotI* and *XbaI* using primers with the N-terminal containing the Kozak and HA sequences (TTACTTCAGGCGGCCGCGCCAAAATGTACCCATACGATGTTCCAGATTAC) and the C-terminal containing the myc sequence (TTAAAAACGATTCTATTCTAGTTACAGATCCTCTTCTGAGATGAGTTTTTGTCTTTCTCTCTGATGAAGAGCTTCAATG). To confirm production of the complete coding sequence, it was bidirectionally sequenced using the Sanger method. These plasmids were injected into embryos and integrated into the attP2 landing site via ΦC31 recombination (WellGenetics, Taiwan).

Immunohistochemistry and imaging

Immunohistochemistry was performed as described previously (Sugie et al., 2017 [DOI](#)). Flies were anesthetized on a CO₂ pad, and specimens of the correct genotype were selected. Specimens were briefly washed in ethanol for a few seconds before being transferred to a Petri dish containing phosphate-buffered saline (PBS). Using forceps, the proboscis was secured, and the right and left eyes were dissected out with another set of forceps. The proboscis and attached trachea were also removed from the brain. The brains were then carefully transferred with forceps into a 1.5 mL tube containing 150 μL of 0.3% PBT (PBS with Triton-X100) at room temperature.

To fix the tissues, 50 μL of 16% formaldehyde was added to each tube, followed by gentle mixing. The samples were incubated in the fixative for 50 minutes at room temperature. Subsequent washing involved three rinses with 200 μL of 0.3% PBT, using a micropipette for fluid exchange. After the final wash, the PBT was removed, and the samples were incubated in the primary antibody solution overnight (O/N) at 4 °C. This was followed by three washes with 0.3% PBT, after which the secondary antibody solution (diluted in 0.3% PBT) was added, and samples were incubated overnight in the dark at 4 °C. After another three washes in 0.3% PBT, the samples were prepared for imaging. The following antibodies were used: rat anti-HA (3F10, 1:400; Roche, Switzerland), mouse anti-Chaoptin (24B10, 1:25; Developmental Studies Hybridoma Bank, USA), rat anti-CadN (DN-Ex#8, 1:50; Developmental Studies Hybridoma Bank), anti-rat Alexa Fluor 488 (1:400; Thermo Fisher Scientific, USA), anti-mouse Alexa Fluor 488 (1:400; Thermo Fisher Scientific), anti-mouse Alexa Fluor 568 (1:400; Thermo Fisher Scientific), and anti-rat Alexa Fluor 633 (1:400; Thermo Fisher Scientific). Specimens were mounted using a Vectashield mounting medium (Vector Laboratories, USA). Images were captured using an FV3000 confocal microscope (Olympus, Japan) or a C2 (Nikon, Japan) confocal microscope for **Figure S1** [DOI](#) and processed using the IMARIS software (Oxford Instruments, UK) or Fiji software, an open-source image analysis software (Schindelin et al., 2012 [DOI](#)).

Electron microscopy

Fly heads were fixed overnight with 2.5% glutaraldehyde and 2% paraformaldehyde in 0.1 M sodium cacodylate buffer. Heads were then post-fixed with 1% osmium tetroxide, dehydrated in ethanol, and embedded in Epon. Ultra-thin tangential sections of the laminas (70 nm) were stained

with uranyl acetate and lead citrate. Random retinas or lamina sections were imaged at a magnification of $\times 100$ k with a VELETA CCD Camera (Olympus) mounted on a JEM 1010 transmission electron microscope (JEOL, Japan). In the R1–6 axons of the lamina, the areas corresponding to mitochondria were quantified using the freehand line tool in NIS Elements software (Nikon). Concurrently, mitochondria were manually categorized based on their structural characteristics: those with a densely packed matrix were classified as Class A, and those with a collapsed structure as Class B.

Quantification of Mito-GFP intensity in retinal axons and retina

To measure the relative intensity of Mito-GFP in the photoreceptor R7/8 axons of the medulla, the imaging analysis software IMARIS was used. First, the M1–6 layers of the medulla, including the R7/8 axons, were manually covered using the surface function. Then, the surface area was masked, the surface of the left axons was stained with anti-Chaoptin, and the Mito-GFP signal was generated automatically using the surface function. Finally, the voxel number of the Mito-GFP surface was divided by the voxel number of the R axon surface. These quantifications were performed by experimenters blind to the genotype.

Mitochondrial superoxide detection and quantification

The ROS level was determined using the MitoSOX Red Mitochondrial Superoxide Indicator (Thermo Fisher Scientific). The procedure involved dissecting adult brains in phosphate-buffered saline (PBS), transferring them to 5 μ M MitoSOX solution, and incubating them for 10 min in the dark at room temperature. The brains were then washed with PBS containing 20% Vectashield (Vector Laboratories), followed by sequential washes with PBS containing 40% and 60% Vectashield. The samples were mounted in Vectashield and scanned using an FV3000 confocal microscope (Olympus). To quantify the ROS level in photoreceptor R7/8 axons, the IMARIS software (Bitplane) was used for imaging analysis. The software generated automatic surfaces of the MitoSOX and Mito-GFP signals. Then, the ROS level was calculated by dividing the voxel number of the MitoSOX surface on the Mito-GFP surface by the voxel number of the Mito-GFP surface. These quantifications were performed by experimenters blind to the genotype.

Measurement of mitophagy levels in retinal axons

The mitophagy levels in the photoreceptor R7/8 axons of the medulla were analyzed using the mitoQC reporter. Mitolysosomes were identified by referencing the red-only mCherry puncta of mitoQC. The number of mitolysosomes of each voxel image was manually quantified with IMARIS software (Bitplane). These quantifications were performed by experimenters blind to the genotype.

Western blot analysis of the whole *Drosophila* body

We regulated protein expression temporally across the whole body using the *Tub-Gal4* and *Tub-GAL80^{TS}* system. Flies harboring each *hOPA1* variant were maintained at a permissive temperature of 20°C, and upon emergence, females were transferred to a restrictive temperature of 29°C for subsequent experiments. The flies were collected in plastic tubes and sonicated (2 x 30 s) using a Q55 sonicator (Qsonica, Newtown, CT, USA) in lysis buffer (10 mM Tris-HCl, pH 7.5, 150 mM NaCl, 1 mM EDTA, 2% DDM: 10 μ l/mg flies) supplemented with 1:1000 (v/v) Protease Inhibitor Cocktail Set III (Calbiochem, USA). Rat anti-HRP-conjugated-HA (3F10, 1: 3000; proteintech, Germany), mouse anti-myc (9B11, 1: 10000; Cell Signaling Technology, USA), rabbit anti-OPA1 (ab42364, 1: 1000; Abcam, UK), rabbit anti-COXII (1: 100; Murari et al. *JCB* 2020), rabbit anti-Atg8a (ab109364, 1: 8000; Abcam), mouse anti- α -Tubulin (T9026, 1:100,000; Sigma-Aldrich, USA), goat HRP-conjugated anti-mouse IgG (SA00001-1, 1: 10,000; proteintech), and goat HRP-conjugated anti-rabbit IgG (7074S, 1: 5,000; Cell Signaling Technology) antibodies were used for Western blotting.

Pairwise alignment of hOPA1 and dOPA1

The amino acid sequences for hOPA1 and dOPA1 were referenced from NM_015560.3 and NM_166040.2, respectively, and aligned using the EMBOSS Needle (Rice et al., 2000 [↗](#)). The GTPase, Middle, and GED domains in dOPA1 were identified by cross-referencing the previously described amino acid positions of these domains in hOPA1 (Liesa et al., 2009 [↗](#)). The degree of identity between the aligned amino acid sequences was subsequently determined.

Experimental design and statistical analyses

Experimental analyses were performed using Prism 9 (GraphPad Software, USA). For **Figures 1E** [↗](#), 1H, 2C, S2B and S7B, data were analyzed using t tests (and nonparametric tests) and Mann-Whitney tests. For **Figure 1J** [↗](#), Fisher's exact test was used. For **Figures 2H** [↗](#), 3F, 3K, 4D, 5C, 5E, 5F, 6A, 6B, S3B, S4B, S4D, S6C and S6E, differences among multiple groups were examined using the Kruskal-Wallis test followed by the two-stage linear step-up procedure of Benjamini, Krieger, and Yekutieli tests. The null hypothesis was rejected at a 0.05 level of significance. Sample sizes are indicated in each figure. The number of retinal axons and rhabdomeres, the intensity of Mito-GFP and MitoSOX, the mitochondrial area and the classification, and the corrected *P* values are described in the Results section. All data are expressed as the mean ± SEM.

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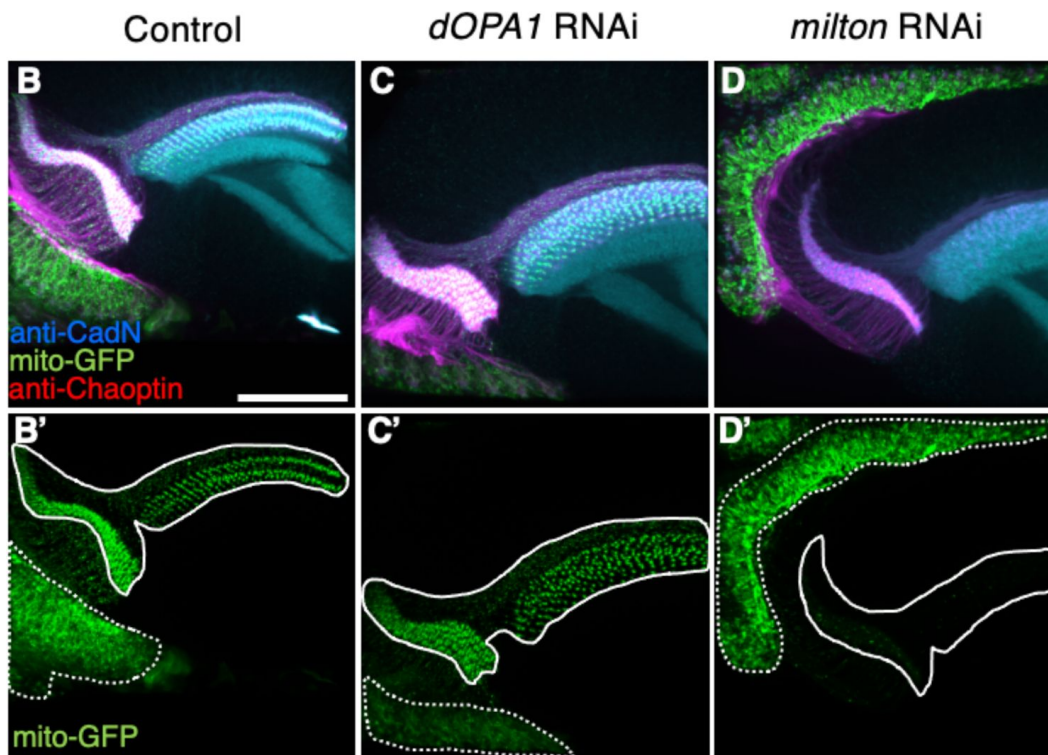
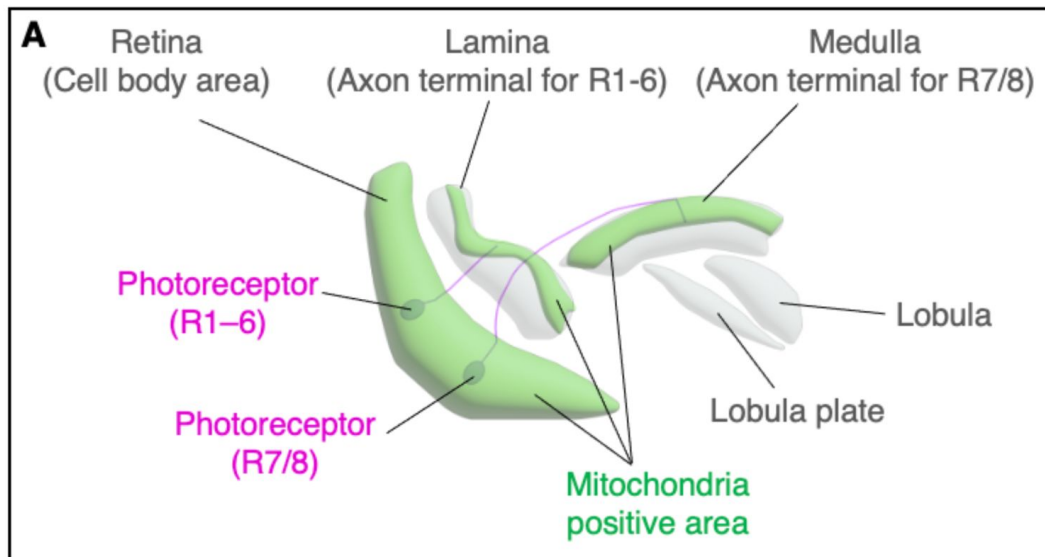


Figure S1.

Analysis of trafficking defects in axons of *dOPA1* knockdown flies.

Schematic illustration of the *Drosophila* visual system around 24 h after puparium formation. The cell bodies of photoreceptors (magenta) are found in the retina; type 1–6 photoreceptors project their axons to the lamina of the primary optic ganglion. The axons of type 7/8 photoreceptors project into the medulla of the secondary optic ganglion. Mitochondria are abundantly located in their cell bodies and axon terminals (green). Higher-order visual centers include the lobula and lobula plate. (B–D) Neuropil and photoreceptor cells were labeled using an anti-CadN antibody (cyan) and anti-Chaoptin antibody (magenta), respectively. Mitochondria were visualized using the expression of Mito-GFP driven by *GMR-Gal4* (green). (B, B') and (C, C') show whole photoreceptors in the RNAi Control and *dOPA1* knockdown, respectively. The regions indicated by dashed and solid lines in (B', C', D') indicate the cell body and axonal areas, respectively. (D, D') shows whole photoreceptors in *milton* knockdown. Scale bar: 50 μ m. See the Table1 for the genotypes of the *Drosophila* used in the experiments.

Figure S2.

The impact on the retinal axons of another RNAi line in *Drosophila* OPA1.

(A) The images show the representations of retinal axons labeled with an anti-Chaoptin antibody (gray). Control and *dOPA1* RNAi (VDRC 106290) from VDRC two weeks after eclosion. Scale bar: 50 μ m. (B) Quantification of the number of axons for each genotype. Control (n = 14 optic lobes) and *dOPA1* RNAi (VDRC 106290) (n = 22 optic lobes). The data are presented as mean $\pm$ SEM. See the Table1 for the genotypes of the *Drosophila* used in the experiments.

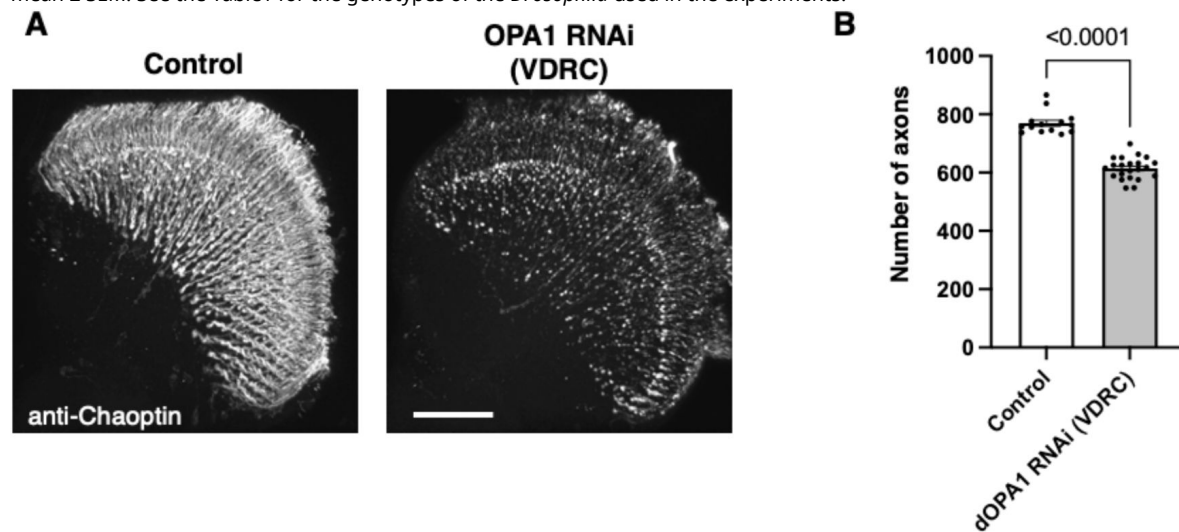
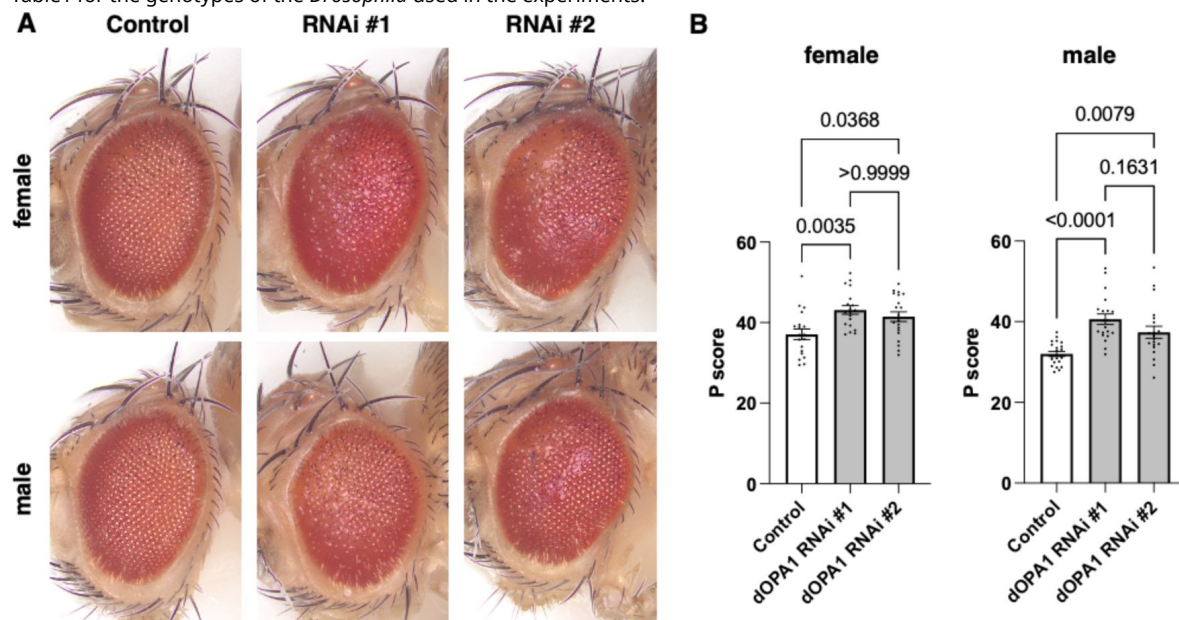


Figure S3.

Evaluation of the impact on *Drosophila* compound eyes due to knockdown of *dOPA1*.

(A) Bright-field microscopic images of representative fly eyes expressing 40D-UAS (control), RNAi #1 (BDSC 32358), and RNAi #2 (VDRC 106290) using the *GMR-Gal4* driver. (B) Graph representing the P score of each genotype: female control (n = 19), female RNAi #1 (n = 19), female RNAi #2 (n = 20), male control (n = 21), male RNAi #1 (n = 20), and male RNAi #2 (n = 20). See the Table1 for the genotypes of the *Drosophila* used in the experiments.



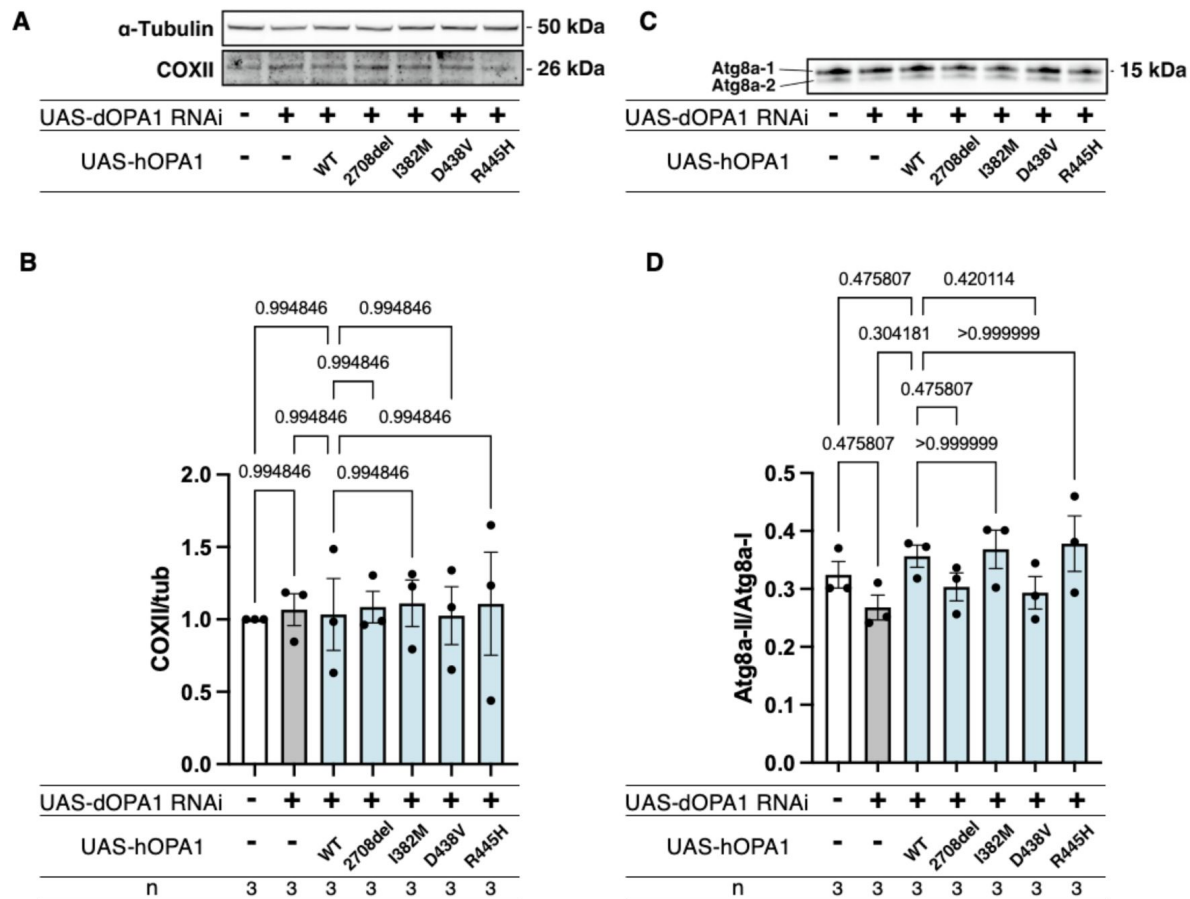


Figure S4.

Effects of *hOPA1* variants on COXII expression and Autophagy in the context of *dOPA1* knockdown.

(A) Western blot analysis to show the expression of COXII upon expression of *hOPA1* variants in the context of *dOPA1* knockdown. (B) Quantification of COXII protein levels. Control (n = 3), *luciferase* RNAi with *dOPA1* RNAi (n = 3), *hOPA1_WT* with *dOPA1* RNAi (n = 3), *hOPA1_2708-2711del* with *dOPA1* RNAi (n = 3), *hOPA1_I382M* with *dOPA1* RNAi (n = 3), *hOPA1_D438V* with *dOPA1* RNAi (n = 3), and *hOPA1_R445H* with *dOPA1* RNAi (n = 3). Data are presented as mean ± SEM. (C) Western blot analysis to show the expression of Atg8a-1 and Atg8a-2 upon expression of *hOPA1* variants in the context of *dOPA1* knockdown. (D) Quantification of the ratio of Atg8a-2 and Atg8a-1. Control (n = 3), *luciferase* RNAi with *dOPA1* RNAi (n = 3), *hOPA1_WT* with *dOPA1* RNAi (n = 3), *hOPA1_2708-2711del* with *dOPA1* RNAi (n = 3), *hOPA1_I382M* with *dOPA1* RNAi (n = 3), *hOPA1_D438V* with *dOPA1* RNAi (n = 3), and *hOPA1_R445H* with *dOPA1* RNAi (n = 3). Data are presented as mean ± SEM.

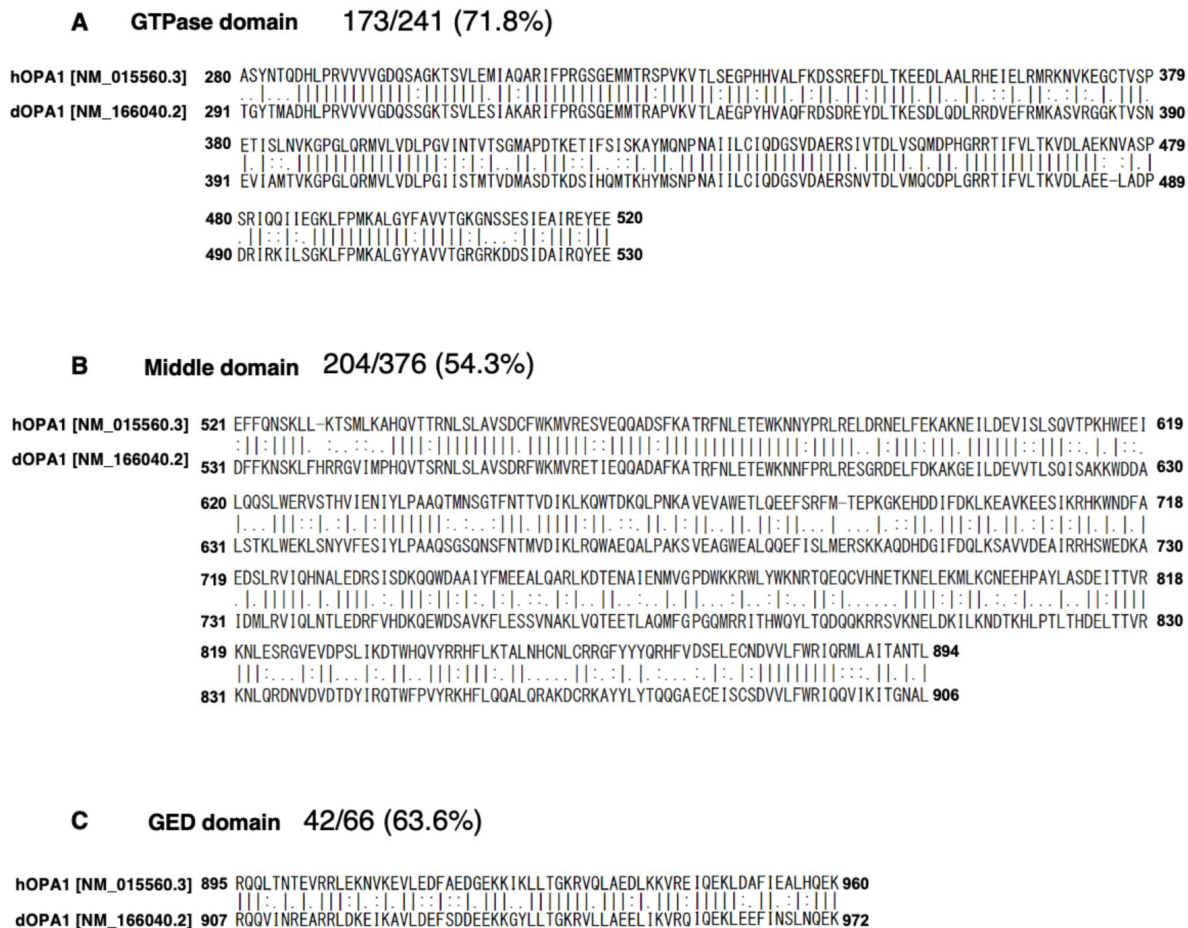


Figure S5.

Pairwise Alignment of GTPase, Middle, and GED Domains of hOPA1 and dOPA1.

(A–C) The amino acid sequences of hOPA1 ([NM_015560.3]) and dOPA1 ([NM_166040.2]) were aligned using EMBOSS Needle. GTPase domain (A), Middle domain (B), and GED domain (C). Each percentage value represents the degree of identity within the amino acid sequence. GTPase domain (280–520), Middle domain (521–894), and GED domain (895–907). The beginning and end of each domain were reported previously (Liesa et al., 2009).

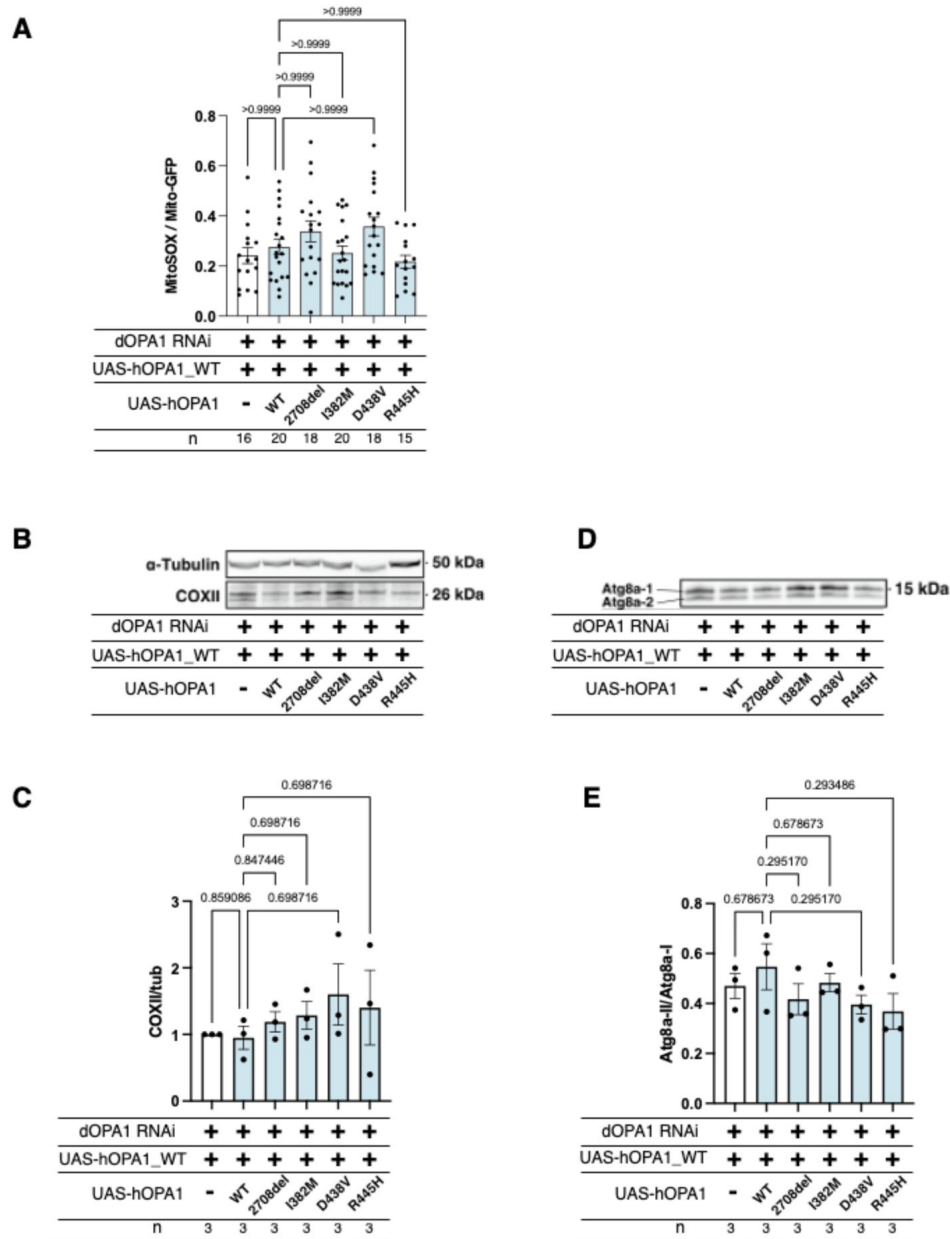


Figure S6.

Comparative analysis of mitochondrial ROS levels, COXII expression, and Autophagy between *hOPA1* wild-type and the variants expression under *dOPA1* knockdown conditions.

(A) Impact on mitochondrial ROS levels upon expression of wild-type *hOPA1* and its variants under conditions of *dOPA1* knockdown. Control (n = 16 optic lobes), *hOPA1_WT* (n = 20 optic lobes), *hOPA1_2708-2711del* (n = 18 optic lobes), *hOPA1_I382M* (n = 20 optic lobes), *hOPA1_D438V* (n = 18 optic lobes), and *hOPA1_R445H* (n = 15 optic lobes). Data are presented as mean $\pm$ SEM. (B) Western blot analysis to show the expression of COXII upon expression of wild-type *hOPA1* and its variants under condition of *dOPA1* knockdown. (C) Quantification of COXII protein levels. Control (n = 3), *hOPA1_WT* (n = 3), *hOPA1_2708-2711del* (n = 3), *hOPA1_I382M* (n = 3), *hOPA1_D438V* (n = 3), and *hOPA1_R445H* (n = 3). Data are presented as mean $\pm$ SEM. (D) Western blot analysis to show the expression of Atg8a-1 and Atg8a-2 upon expression of wild-type *hOPA1* and its variants under condition of *dOPA1* knockdown. (E) Quantification of the ratio of Atg8a-2 and Atg8a-1. Control (n = 3), *hOPA1_WT* (n = 3), *hOPA1_2708-2711del* (n = 3), *hOPA1_I382M* (n = 3), *hOPA1_D438V* (n = 3), and *hOPA1_R445H* (n = 3). Data are presented as mean $\pm$ SEM. See the Table1 for the genotypes of the *Drosophila* used in the experiments.

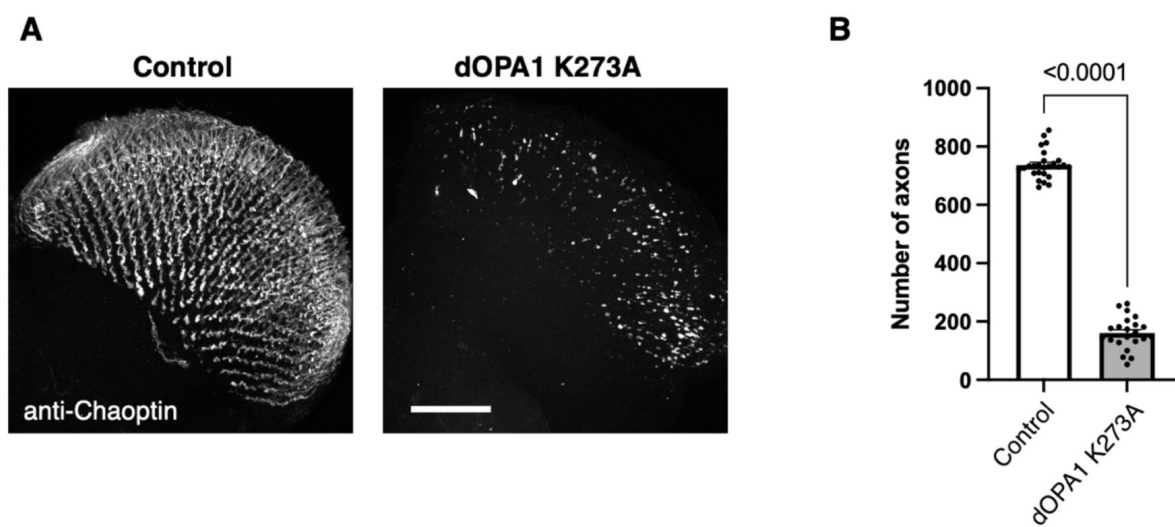


Figure S7.

The effect of a dominant-negative mutation in *dOPA1*.

(A) The images show the representations of retinal axons labeled with an anti-Chaoptin antibody (gray). Control and the expression of a presumably GTPase-negative form of *dOPA1*^{K273A} 1 week after eclosion. Scale bar: 50 μ m. (B) Quantification of the number of axons for each genotype. Control (n = 26 optic lobes) and *dOPA1*^{K273A} (n = 22 optic lobes). The data are presented as mean $\pm$ SEM. See the Table1 for the genotypes of the *Drosophila* used in the experiments.

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Editors

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Reviewer #3 (Public Review):

Nitta *et al.* use a fly model of autosomal dominant optic atrophy to provide mechanistic insights into distinct disease-causing OPA1 variants. It has long been hypothesized that missense OPA1 mutations affecting the GTPase domain, which are associated with more severe optic atrophy and extra-ophthalmic neurologic conditions such as sensorineural hearing loss (DOA plus), impart their effects through a dominant negative mechanism, but no clear direct evidence for this exists particularly in an animal model. The authors execute a well-designed study to establish their model, demonstrating a mitochondrial phenotype and optic atrophy measured as axonal degeneration. They leverage this model to provide the first direct evidence for a dominant negative mechanism for 2 mutations causing DOA plus by expressing these variants in the background of a full hOPA1 complement.

Strengths of the paper include well-motivated objectives and hypotheses, and overall solid design and execution. There is a thorough discussion of the interpretation and context of the findings. The results technically support their primary conclusions with minor limitations. First, while only partial rescue of the most clinically relevant metric for optic atrophy in this model is now acknowledged, the result nevertheless hamstrings the mechanistic experiments that follow. Second, the results statistically support a dominant negative effect of DOA plus-associated variants, yet the data show a marginal impact on axonal degeneration for these variants. In added experiments, the ability of WT hOPA1 and I382M but not 2708del, D438V or R445H to rescue ROS levels or mitophagy in the context of dOPA1 knockdown serves to support axonal number as a valid measure of mitochondrial function in this context. However, the critical experiment demonstrating a dominant negative effect was performed in the context of expressing WT hOPA1 along with a pathogenic variant, in which no differences in ROS, COXII expression or mitophagy were seen. This makes it difficult to

conclude that the dominant negative effect of D438V and R445H on axon number is related to mitochondrial function.

As an animal model of DOA that may serve for rapid assessment of suspected OPA1 variants, the results overall support utility of this model in identifying pathogenic variants but not in distinguishing haploinsufficiency from dominant negative mechanisms among those variants. The impact of this work in providing the first direct evidence of a dominant negative mechanism is under-stated considering how important this question is in development of genetic treatments for dominant optic atrophy.

Comments on revised version:

The authors have addressed the comments in my initial review. Through these modification and those related to the comments from the other reviewers, the manuscript is strengthened.

Comments on author responses to each of the reviews:

Reviewer 1:

Interpretation of data has been appropriately reorganized in the discussion.

Quantified mitochondria in the model show no difference in number. There is reduced size and structural abnormalities on electron microscopy.

Application of mito-QC revealed increased mitophagy.

Regarding partial rescue of axonal number in the mutant model, statistical significance between control and rescue is still not depicted in Figure 4D. Detailing possible explanations for this has been addressed in the discussion. However, only partial rescue of the most clinically relevant metric for optic atrophy in this model hampers subsequent mechanistic experiments that follow.

Discussion regarding variant I382M has been improved.

While reviewer 1's concerns about axonal number as a biomarker for OPA1 function are valid, it is worth noting that this is the most clinically relevant marker in the context of DOA. That said, I agree that the mechanistic DN/HI studies needed support using other measures of mitochondrial function, and the authors have done this. The ability of WT hOPA1 and I382M but not 2708del, D438V or R445H to rescue ROS levels or mitophagy in the context of dOPA1 knockdown serves to support axonal number as a valid measure of mitochondrial function in this context. However, the critical experiment demonstrating a dominant negative effect was performed in the context of expressing WT hOPA1 along with a pathogenic variant, in which no differences in ROS, COXII expression or mitophagy were seen. This makes it difficult to conclude that the (marginal) DN effect of D438V and R445H on axon number is related to mitochondrial function, and serves as a minor weakness of the paper.

Which exons are included in the transcript, and therefore, which isoforms are expressed in the model, has been addressed.

Reviewer 2:

The authors have addressed the need to include greater methodological details.

Language concerning the clinical utility of the model in informing treatment decisions has been appropriately modified. As pointed out by Reviewer 1, additional studies were needed to better establish the potential clinical utility of this model in screening DOA variants. The authors have completed those experiments, and the results overall support utility of this model in identifying pathogenic variants but not in distinguishing HI/DN mechanisms among those variants.

Reviewer 3:

The author has addressed the partial rescue effect as above.

The authors have not modified the text to acknowledge the marginal effect sizes in the critical experiment of the study that demonstrates a DN effect. Statistically, the results indeed support a dominant negative effect of DOA plus-associated variants, yet the data show a marginal impact on axonal degeneration for these variants. This remains a weakness of the study.

<https://doi.org/10.7554/eLife.87880.2.sa1>

Author response:

The following is the response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Nitta et al, in their manuscript titled, "Drosophila model to clarify the pathological significance of OPA1 in autosomal dominant optic atrophy." The novelty of this paper lies in its use of human (hOPA1) to try to rescue the phenotype of an OPA1 +/- Drosophila DOA model (dOPA). The authors then use this model to investigate the differences between dominant-negative and haploinsufficient OPA1 variants. The value of this paper lies in the study of DN/HI variants rather than the establishment of the drosophila model per se as this has existed for some time and does have some significant disadvantages compared to existing models, particularly in the extra-ocular phenotype which is common with some OPA1 variants but not in humans. I judge the findings of this paper to be valuable with regards to significance and solid with regards to the strength of the evidence.

Suggestions for improvements:

(1) Stylistically the results section appears to have significant discussion/conclusion/inferences in section with reference to existing literature. I feel that this information would be better placed in the separate discussion section. E.g. lines 149-154.

We appreciate the reviewer's suggestion to relocate the discussion, conclusions, and inferences, particularly those that reference existing literature, to a separate discussion section. For lines 149–154, we placed them in the discussion section (lines 343–347) as follows. "Our established fly model is the first simple organism to allow observation of degeneration of the retinal axons. The mitochondria in the axons showed fragmentation of mitochondria. Former studies have observed mitochondrial fragmentation in S2 cells (McQuibban et al., 2006), muscle tissue (Deng et al., 2008), segmental nerves (Trevisan et al., 2018), and ommatidia (Yarosh et al., 2008) due to the LOF of *dOPA1*."

For lines 178–181, we also placed them in the discussion section (lines 347–351) as follows. "Our study presents compelling evidence that *dOPA1* knockdown instigates neuronal degeneration, characterized by a sequential deterioration at the axonal terminals and extending to the cell bodies. This degenerative pattern, commencing from the distal axons and progressing proximally towards the cell soma, aligns with the paradigm of 'dying-back' neuropathy, a phenomenon extensively documented in various neurodegenerative disorders (Wang et al., 2012)."

For lines 213–217, 218–220, and 222–223, we also placed them in the discussion section (lines 363– 391) as follows. “To elucidate the pathophysiological implications of mutations in the *OPA1* gene, we engineered and expressed several human *OPA1* variants, including the 2708-2711del mutation, associated with DOA, and the I382M mutation, located in the GTPase domain and linked to DOA. We also investigated the D438V and R445H mutations in the GTPase domain and correlated with the more severe DOA plus phenotype. The 2708-2711del mutation exhibited limited detectability via HA-tag probing. Still, it was undetectable with a myc tag, likely due to a frameshift event leading to the mutation's characteristic truncated protein product, as delineated in prior studies (Zanna et al., 2008). Contrastingly, the I382M, D438V, and R445H mutations demonstrated expression levels comparable to the WT *hOPA1*. However, the expression of these mutants in retinal axons did not restore the dOPA1 deficiency to the same extent as the WT *hOPA1*, as evidenced in Figure 5E. This finding indicates a functional impairment imparted by these mutations, aligning with established understanding (Zanna et al., 2008). Notably, while the 2708-2711del and I382M mutations exhibited limited functional rescue, the D438V and R445H mutations did not show significant rescue activity. This differential rescue efficiency suggests that the former mutations, particularly the I382M, categorized as a hypomorph (Del Dotto et al., 2018), may retain partial functional capacity, indicative of a LOF effect but with residual activity. The I382M missense mutation within the GTPase domain of *OPA1* has been described as a mild hypomorph or a disease modifier. Intriguingly, this mutation alone does not induce significant clinical outcomes, as evidenced by multiple studies (Schaaf et al., 2011; Bonneau et al., 2014; Bonifert et al., 2014; Carelli et al., 2015). A significant reduction in protein levels has been observed in fibroblasts originating from patients harboring the I382M mutation. However, mitochondrial volume remains unaffected, and the fusion activity of mitochondria is only minimally influenced (Kane et al., 2017; Del Dotto et al., 2018). This observation is consistent with findings reported by de la Barca et al. in Human Molecular Genetics 2020, where a targeted metabolomics approach classified I382M as a mild hypomorph. In our current study, the I382M mutation preserves more *OPA1* function compared to DN mutations, as depicted in Figures 5E and F. Considering the results from our *Drosophila* model and previous research, we hypothesize that the I382M mutation may constitute a mild hypomorphic variant. This might explain its failure to manifest a phenotype on its own, yet its contribution to increased severity when it occurs in compound heterozygosity.

(2) I do think further investigation as to why a reduction of mitochondria was noticed in the knockdown. There are conflicting reports on this in the literature. My own experience of this is fairly uniform mitochondrial number in WT vs OPA1 variant lines but with an increased level of mitophagy presumably reflecting a greater turnover. There are a number of ways to quantify mitochondrial load e.g. mtDNA quantification, protein quantification for tom20/hsp60 or equivalent. I feel the reliance on ICC here is not enough to draw conclusions. Furthermore, mitophagy markers could be checked at the same time either at the transcript or protein level. I feel this is important as it helps validate the drosophila model as we already have a lot of experimental data about the number and function of mitochondria in OPA+/- human/mammalian cells.

We thank the reviewer for the insightful comments and suggestions regarding our study on the impact of mitochondrial reduction in a knockdown model. We concur with the reviewer's observation that our initial results did not definitively demonstrate a decrease in the number of mitochondria in retinal axons. Furthermore, we measured mitochondrial quantity by conducting western blotting using antiCOXII and found no reduction in mitochondrial content with the knockdown of *dOPA1* (Figure S4A and B). Consequently, we have revised our manuscript to remove the statement “suggesting a decreased number of mitochondria in retinal axons. However, whether this decrease is due to degradation resulting from a decline in mitochondrial quality or axonal transport failure remains unclear.” Instead, we have

refocused our conclusion to reflect our electron microscopy findings, which indicate reduced mitochondrial size and structural abnormalities. The reviewer's observation of consistent mitochondrial numbers in WT versus mutant variant lines and elevated mitophagy levels prompted us to evaluate mitochondrial turnover as a significant factor in our study. Regarding verifying mitophagy markers, we incorporated the mito-QC marker in our experimental design. In our experiments, mito-QC was expressed in the retinal axons of *Drosophila* to assess mitophagy activity upon *dOPA1* knockdown. We observed a notable increase in mCherry positive but GFP negative puncta signals one week after eclosion, indicating the activation of mitophagy (Figure 2D–H). This outcome strongly suggests that *dOPA1* knockdown enhances mitophagy in our *Drosophila* model. The application of mito-QC as a quantitative marker for mitophagy, validated in previous studies, offers a robust approach to analyzing this process. Our findings elucidate the role of *dOPA1* in mitochondrial dynamics and its implications for neuronal health. These results have been incorporated into Figure 2, with the corresponding text updated as follows (lines 159–167): “Given that an increase in mitophagy activity has been reported in mouse RGCs and nematode ADOA models (Zaninello et al., 2022; Zaninello et al., 2020), the mitoQC marker, an established indicator of mitophagy activity, was expressed in the photoreceptors of *Drosophila*. The mito-QC reporter consists of a tandem mCherry-GFP tag that localizes to the outer membrane of mitochondria (Lee et al., 2018). This construct allows the measurement of mitophagy by detecting an increase in the red-only mCherry signal when the GFP is degraded after mitochondria are transported to lysosomes. Post *dOPA1* knockdown, we observed a significant elevation in mCherry positive and GFP negative puncta signals at one week, demonstrating an activation of mitophagy as a consequence of *dOPA1* knockdown (Figure 2D–H).”

(3) Could the authors comment on the failure of the *dOPA1* rescue to return their biomarker, axonal number to control levels. In Figure 4D is there significance between the control and rescue. Presumably so as there is between the mutant and rescue and the difference looks less.

As the reviewer correctly pointed out, there is a significant difference between the control and rescue groups, which we have now included in the figure. Additionally, we have incorporated the following comments in the discussion section (lines 329–342) regarding this significant difference: “In our study, expressing *dOPA1* in the retinal axons of *dOPA1* mutants resulted in significant rescue, but it did not return to control levels. There are three possible explanations for this result. The first concerns gene expression levels. The Gal4-line used for the rescue experiments may not replicate the expression levels or timing of endogenous *dOPA1*. Considering that the optimal functionality of *dOPA1* may be contingent upon specific gene expression levels, attaining a wild-type-like state necessitates the precise regulation of these expression levels. The second is a nonautonomous issue. Although *dOPA1* gene expression was induced in the retinal axons for the rescue experiments, many retinal axons were homozygous mutants, while other cell types were heterozygous for the *dOPA1* mutation. If there is a non-autonomous effect of *dOPA1* in cells other than retinal axons, it might not be possible to restore the wild-type-like state fully. The third potential issue is that only one isoform of *dOPA1* was expressed. In mouse OPA1, to completely restore mitochondrial network shape, an appropriate balance of at least two different isoforms, lOPA1 and s-OPA1, is required (Del Dotto et al., 2017). This requirement implies that multiple isoforms of *dOPA1* are essential for the dynamic activities of mitochondria.”

(4) The authors have chosen an interesting if complicated missense variant to study, namely the I382M with several studies showing this is insufficient to cause disease in isolation and appears in high frequency on gnomAD but appears to worsen the phenotype when it appears as a compound het. I think this is worth discussing in the context of the results, particularly with regard to the ability for this variant to partially rescue the *dOPA1* model as shown in Figure 5.

As the reviewer pointed out, the I382M mutation is known to act as a disease modifier. However, in our system, as suggested by Figure 5, I382M appears to retain more activity than DN mutations. Considering previous studies, we propose that I382M represents a mild hypomorph. Consequently, while I382M alone may not exhibit a phenotype, it could exacerbate severity in a compound heterozygous state. We have incorporated this perspective in our revised discussion (lines 375-391).

“Notably, while the 2708-2711del and I382M mutations exhibited limited functional rescue, the D438V and R445H mutations did not show significant rescue activity. This differential rescue efficiency suggests that the former mutations, particularly the I382M, categorized as a hypomorph (Del Dotto et al., 2018), may retain partial functional capacity, indicative of a LOF effect but with residual activity. The I382M missense mutation within the GTPase domain of OPA1 has been described as a mild hypomorph or a disease modifier. Intriguingly, this mutation alone does not induce significant clinical outcomes, as evidenced by multiple studies (Schaaf et al., 2011; Bonneau et al., 2014; Bonifert et al., 2014; Carelli et al., 2015). A significant reduction in protein levels has been observed in fibroblasts originating from patients harboring the I382M mutation. However, mitochondrial volume remains unaffected, and the fusion activity of mitochondria is only minimally influenced (Kane et al., 2017; Del Dotto et al., 2018). This observation is consistent with findings reported by de la Barca et al. in Human Molecular Genetics 2020, where a targeted metabolomics approach classified I382M as a mild hypomorph. In our current study, the I382M mutation preserves more OPA1 function compared to DN mutations, as depicted in Figures 5E and F. Considering the results from our *Drosophila* model and previous research, we hypothesize that the I382M mutation may constitute a mild hypomorphic variant. This might explain its failure to manifest a phenotype on its own, yet its contribution to increased severity when it occurs in compound heterozygosity.”

(5) I feel the main limitation of this paper is the reliance on axonal number as a biomarker for OPA1 function and ultimately rescue. I have concerns because a) this is not a well validated biomarker within the context of OPA1 variants b) we have little understanding of how this is affected by over/under expression and c) if it is a threshold effect e.g. once OPA1 levels reach <x% pathology develops but develops normally when opa1 expression is >x%. I think this is particularly relevant when the authors are using this model to make conclusions on dominant negativity/HI with the authors proposing that if expression of a hOPA1 transcript does not increase opa1 expression in a dOPA1 KO then this means that the variant is DN. The authors have used other biomarkers in parts of this manuscript e.g. ROS measurement and mito trafficking but I feel this would benefit from something else particularly in the latter experiments demonstrated in figure 5 and 6.

The reviewer raised concerns regarding the adequacy of axonal count as a validated biomarker in the context of *OPA1* mutants. In response, we corroborated its validity using markers such as MitoSOX, Atg8, and COXII. Experiments employing MitoSOX revealed that the augmented ROS signals resulting from *dOPA1* knockdown were mitigated by expressing human *OPA1*. Conversely, the mutant variants 2708-2711del, D438V, and R445H did not ameliorate these effects, paralleling the phenotype of axonal degeneration observed. These findings are documented in Figure 5F, and we have incorporated the following text into section lines 248–254 of the results:

“Furthermore, we assessed the potential for rescuing ROS signals. Similar to its effect on axonal degeneration, wild-type hOPA1 effectively mitigated the phenotype, whereas the 2708-2711del, D438V, and R445H mutants did not (Figure 5F). Importantly, the I382M variant also reduced ROS levels comparably to the wild type. These findings demonstrate that both axonal degeneration and the increase in ROS caused by *dOPA1* downregulation can be effectively

counteracted by hOPA1. Although I382M retains partial functionality, it acts as a relatively weak hypomorph in this experimental setup.”

Moreover, utilizing mito-QC, we observed elevated mitophagy in our *Drosophila* model, with these results now included in Figure 2D–H. Given the complexity of the genetics involved and the challenges in establishing lines, autophagy activity was quantified by comparing the ratio of Atg8-1 to Atg8-2 via Western blot analysis. However, no significant alterations were detected across any of the genotypes. Additionally, mitochondrial protein levels derived from COXII confirmed consistent mitochondrial quantities, showing no considerable variance following knockdown. These insights affirm that retinal axon degeneration and mitophagy activation are present in the *Drosophila* DOA model, although the Western blot analysis revealed no significant changes in autophagy activation. Such findings necessitate caution as this model may not fully replicate the molecular pathology of the corresponding human disease. These Western blot findings are presented in Figure S4, with the following addition made to section lines 255–263 of the results:

“We also conducted Western blot analyses using anti-COXII and anti-Atg8a antibodies to assess changes in mitochondrial quantity and autophagy activity following the knockdown of *dOPA1*. Mitochondrial protein levels, indicated by COXII quantification, were evaluated to verify mitochondrial content, and the ratio of Atg8a-1 to Atg8a-2 was used to measure autophagy activation. For these experiments, *Tub-Gal4* was employed to systemically knockdown *dOPA1*. Considering the lethality of a whole-body *dOPA1* knockdown, *Tub-Gal80TS* was utilized to repress the knockdown until eclosion by maintaining the flies at 20°C. After eclosion, we increased the temperature to 29°C for two weeks to induce the knockdown or expression of *hOPA1* variants. The results revealed no significant differences across the genotypes tested (Figure S4A–D).”

In assessing the effects of dominant negative mutations, measurements including ROS levels, the ratio of Atg8-1 to Atg8-2, and the quantity of COXII protein were conducted, yet no significant differences were observed (Figure S6). This limitation of the fly model is mentioned in the results, noting the observation of the axonal degeneration phenotype but not alterations in ROS signaling, autophagy activity, or mitochondrial quantity as follows (line 287–290):

“We investigated the impacts of dominant negative mutations on mitochondrial oxidation levels, mitochondrial quantity, and autophagy activation levels; however, none of these parameters showed statistical significance (Figure S6).”

The reviewer also inquired about the effects of overexpressing and underexpressing *OPA1* on axonal count and whether these effects are subject to a threshold. In response, we expressed both wild-type and variant forms of human *OPA1* in *Drosophila* in vivo and assessed their protein levels using Western blot analysis. The results showed no significant differences in expression levels between the wild-type and variant forms in the *OPA1* overexpression experiments, suggesting the absence of a variation threshold effect. These findings have been newly documented as quantitative data in Figure 5C. Furthermore, we have included a statement in the results section for Figure 6A, clarifying that overexpression of hOPA1 exhibited no discernible impact, as detailed on lines 274–276.

“The results presented in Figure 5C indicate that there are no significant differences in the expression levels among the variants, suggesting that variations in expression levels do not influence the outcomes.”

(6) Could the authors clarify what exons in Figure 5 are included in their transcript. My understanding is transcript NM_015560.3 contains exon 4,4b but not 5b. According to Song 2007 this transcript produces invariably s-OPA1 as it contains the exon 4b cleavage site. If this is true, this is a critical limitation in this study and in my opinion significantly

undermines the likelihood of the proposed explanation of the findings presented in Figure 6. The primarily functional location of OPA1 is at the IMM and l-OPA1 is the primary opa1 isoform probably only that localizes here as the additional AA act as a IMM anchor. Given this is where GTPase likely oligomerizes the expression of s-OPA1 only is unlikely to interact anyway with native protein. I am not aware of any evidence s-OPA1 is involved in oligomerization. Therefore I don't think this method and specifically expression of a hOPA1 transcript which only makes s-OPA1 to be a reliable indicator of dominant negativity/interference with WT protein function. This could be checked by blotting UAS-hOPA1 protein with a OPA1 antibody specific to human OPA1 only and not to dOPA1. There are several available on the market and if the authors see only s-OPA1 then it confirms they are not expressing l-OPA1 with their hOPA1 construct.

As suggested by the reviewer, we performed a Western blot using a human OPA1 antibody to determine if the expressed hOPA1 was producing the l-OPA1 isoform, as shown in band 2 of Figure 5D. The results confirmed the presence of both l-OPA1 and what appears to be s-OPA1 in bands 2 and 4, respectively. These findings are documented in the updated Figure 5D, with a detailed description provided in the manuscript at lines 224-226. Additionally, the NM_015560.3 refers to isoform 1, which includes only exons 4 and 5, excluding exons 4b and 5b. This isoform can express both l-OPA1 and s-OPA1 (refer to Figure 1 in Song et al., J Cell Biol. 2007). We have updated the schematic diagram in the figure to include these exons. The formation of s-OPA1 through cleavage occurs at the OMA1 target site located in exon 5 and the Yme1L target site in exon 5b of OPA1. Isoform 1 of OPA1 is prone to cleavage by OMA1, but a homologous gene for OMA1 does not exist in *Drosophila*. Although a homologous gene for Yme1L is present in *Drosophila*, exon 5b is missing in isoform 1 of OPA1, leaving the origin of the smaller band resembling s-OPA1 unclear at this point.

Reviewer #2 (Public Review):

*The data presented support and extend some previously published data using *Drosophila* as a model to unravel the cellular and genetic basis of human Autosomal dominant optic atrophy (DOA). In human, mutations in OPA1, a mitochondrial dynamin like GTPase (amongst others), are the most common cause for DOA. By using a *Drosophila* loss-of-function mutations, RNAi- mediated knockdown and overexpression, the authors could recapitulate some aspects of the disease phenotype, which could be rescued by the wild-type version of the human gene. Their assays allowed them to distinguish between mutations causing human DOA, affecting the optic system and supposed to be loss-of-function mutations, and those mutations supposed to act as dominant negative, resulting in DOA plus, in which other tissues/organs are affected as well. Based on the lack of information in the Materials and Methods section and in several figure legends, it was not in all cases possible to follow the conclusions of the authors.*

We appreciate the reviewer's constructive feedback and the emphasis on enhancing clarity in our manuscript. We recognize the concerns raised about the lack of detailed information in the Materials and Methods section and several figure legends, which may have obscured our conclusions. In response, we have appended the detailed genotypes of the *Drosophila* strains used in each experiment to a supplementary table. Additionally, we realized that the description of 'immunohistochemistry and imaging' was too brief, previously referenced simply as "immunohistochemistry was performed as described previously (Sugie et al., 2017)." We have now expanded this section to include comprehensive methodological details. Furthermore, we have revised the figure legends to provide clearer and more thorough descriptions.

Similarly, how the knowledge gained could help to "inform early treatment decisions in patients with mutations in hOPA1" (line 38) cannot be followed.

To address the reviewer's comments, we have refined our explanation of the clinical relevance of our findings as follows. We believe this revision succinctly articulates the practical application of our research, directly responding to the reviewer's concerns about linking the study's outcomes to treatment decisions for patients with hOPA1 mutations. By underscoring the model's value in differential diagnosis and its influence on initiating treatment strategies, we have clarified this connection explicitly, within the constraints of the abstract's word limit. The revised sentence now reads: "This fly model aids in distinguishing DOA from DOA plus and guides initial hOPA1 mutation treatment strategies."

Reviewer #3 (Public Review):

Nitta et al. establish a fly model of autosomal dominant optic atrophy, of which hundreds of different OPA1 mutations are the cause with wide phenotypic variance. It has long been hypothesized that missense OPA1 mutations affecting the GTPase domain, which are associated with more severe optic atrophy and extra-ophthalmic neurologic conditions such as sensorineural hearing loss (DOA plus), impart their effects through a dominant negative mechanism, but no clear direct evidence for this exists particularly in an animal model. The authors execute a well-designed study to establish their model, demonstrating a clear mitochondrial phenotype with multiple clinical analogs including optic atrophy measured as axonal degeneration. They then show that hOPA1 mitigates optic atrophy with the same efficacy as dOPA1, setting up the utility of their model to test disease-causing hOPA1 variants. Finally, they leverage this model to provide the first direct evidence for a dominant negative mechanism for 2 mutations causing DOA plus by expressing these variants in the background of a full hOPA1 complement.

Strengths of the paper include well-motivated objectives and hypotheses, overall solid design and execution, and a generally clear and thorough interpretation of their results. The results technically support their primary conclusions with caveats. The first is that both dOPA1 and hOPA1 fail to fully restore optic axonal integrity, yet the authors fail to acknowledge that this only constitutes a partial rescue, nor do they discuss how this fact might influence our interpretation of their subsequent results.

As the reviewer rightly points out, neither dOPA1 nor hOPA1 achieve a complete recovery. Therefore, we acknowledge that this represents only a partial rescue and have added the following explanations regarding this partial rescue in the results and discussion sections.

Result:

Significantly —> partially (lines 207 and 228) Discussion (lines 329–342):

In our study, expressing dOPA1 in the retinal axons of dOPA1 mutants resulted in significant rescue, but it did not return to control levels. There are three possible explanations for this result. The first concerns gene expression levels. The Gal4-line used for the rescue experiments may not replicate the expression levels or timing of endogenous dOPA1. Considering that the optimal functionality of dOPA1 may be contingent upon specific gene expression levels, attaining a wild-type-like state necessitates the precise regulation of these expression levels. The second is a non-autonomous issue. Although dOPA1 gene expression was induced in the retinal axons for the rescue experiments, many retinal axons were homozygous mutants, while other cell types were heterozygous for the dOPA1 mutation. If there is a non-autonomous effect of dOPA1 in cells other than retinal axons, it might not be possible to restore the wild-type-like state fully. The third potential issue is that only one isoform of dOPA1 was expressed. In mouse OPA1, to completely restore mitochondrial

network shape, an appropriate balance of at least two different isoforms, l-OPA1 and s-OPA1, is required (Del Dotto et al., 2017). This requirement implies that multiple isoforms of *dOPA1* are essential for the dynamic activities of mitochondria.

The second caveat is that their effect sizes are small. Statistically, the results indeed support a dominant negative effect of DOA plus-associated variants, yet the data show a marginal impact on axonal degeneration for these variants. The authors might have considered exploring the impact of these variants on other mitochondrial outcome measures they established earlier on. They might also consider providing some functional context for this marginal difference in axonal optic nerve degeneration.

In response to the reviewer's comment regarding the modest effect sizes observed, we acknowledge that the magnitude of the reported changes is indeed small. To explore the impact of these variants on additional mitochondrial outcomes as suggested, we employed markers such as MitoSOX, Atg8, and COXII for validation. However, we could not detect any significant effects of the DOA plus-associated variants using these methods. We apologize for the redundancy, but to address Reviewer #1's fifth question, we present experimental results showing that while the increased ROS signals observed upon *dOPA1* knockdown were rescued by expressing human *OPA1*, the mutant variants 2708-2711del, D438V, and R445H did not ameliorate this effect. This outcome mirrors the axonal degeneration phenotype and is documented in Figure 5F. The following text has been added to the results section lines 248–254:

“Furthermore, we assessed the potential for rescuing ROS signals. Similar to its effect on axonal degeneration, wild-type hOPA1 effectively mitigated the phenotype, whereas the 2708-2711del, D438V, and R445H mutants did not (Figure 5F). Importantly, the I382M variant also reduced ROS levels comparably to the wild type. These findings demonstrate that both axonal degeneration and the increase in ROS caused by *dOPA1* downregulation can be effectively counteracted by hOPA1. Although I382M retains partial functionality, it acts as a relatively weak hypomorph in this experimental setup.”

Moreover, utilizing mito-QC, we observed elevated mitophagy in our *Drosophila* model, with these results now included in Figure 2D–H. Given the complexity of the genetics involved and the challenges in establishing lines, autophagy activity was quantified by comparing the ratio of Atg8-1 to Atg8-2 via Western blot analysis. However, no significant alterations were detected across any of the genotypes. Additionally, mitochondrial protein levels derived from COXII confirmed consistent mitochondrial quantities, showing no considerable variance following knockdown. These insights affirm that retinal axon degeneration and mitophagy activation are present in the *Drosophila* DOA model, although the Western blot analysis revealed no significant changes in autophagy activation. Such findings necessitate caution as this model may not fully replicate the molecular pathology of the corresponding human disease. These Western blot findings are presented in Figure S4, with the following addition made to section lines 255–263 of the results:

“We also conducted Western blot analyses using anti-COXII and anti-Atg8a antibodies to assess changes in mitochondrial quantity and autophagy activity following the knockdown of *dOPA1*. Mitochondrial protein levels, indicated by COXII quantification, were evaluated to verify mitochondrial content, and the ratio of Atg8a-1 to Atg8a-2 was used to measure autophagy activation. For these experiments, *Tub-Gal4* was employed to systemically knockdown *dOPA1*. Considering the lethality of a whole-body *dOPA1* knockdown, *Tub-Gal80TS* was utilized to repress the knockdown until eclosion by maintaining the flies at 20°C. After eclosion, we increased the temperature to 29°C for two weeks to induce the knockdown or expression of *hOPA1* variants. The results revealed no significant differences across the genotypes tested (Figure S4A–D).”

In assessing the effects of dominant negative mutations, measurements including ROS levels, the ratio of Atg8-1 to Atg8-2, and the quantity of COXII protein were conducted, yet no significant differences were observed (Figure S6). This limitation of the fly model is mentioned in the results, noting the observation of the axonal degeneration phenotype but not alterations in ROS signaling, autophagy activity, or mitochondrial quantity as follows (line 287–290):

“We investigated the impacts of dominant negative mutations on mitochondrial oxidation levels, mitochondrial quantity, and autophagy activation levels; however, none of these parameters showed statistical significance (Figure S6).”

Despite these caveats, the authors provide the first animal model of DOA that also allows for rapid assessment and mechanistic testing of suspected OPA1 variants. The impact of this work in providing the first direct evidence of a dominant negative mechanism is under-stated considering how important this question is in development of genetic treatments for DOA. The authors discuss important points regarding the potential utility of this model in clinical science. Comments on the potential use of this model to investigate variants of unknown significance in clinical diagnosis requires further discussion of whether there is indeed precedent for this in other genetic conditions (since the model is nevertheless so evolutionarily removed from humans).

As suggested by the reviewer, we have expanded the discussion in our study to emphasize in greater detail the significance of the fruit fly model and the MeDUsA software we have developed, elaborating on the model's potential applications in clinical science and its precedents in other genetic disorders. Our text is as follows (lines 299–318):

“We have previously utilized MeDUsA to quantify axonal degeneration, applying this methodology extensively to various neurological disorders. The robust adaptability of this experimental system is demonstrated by its application in exploring a wide spectrum of genetic mutations associated with neurological conditions, highlighting its broad utility in neurogenetic research. We identified a novel de novo variant in Spliceosome Associated Factor 1, Recruiter of U4/U6.U5 Tri-SnRNP (*SART1*). The patient, born at 37 weeks with a birth weight of 2934g, exhibited significant developmental delays, including an inability to support head movement at 7 months, reliance on tube feeding, unresponsiveness to visual stimuli, and development of infantile spasms with hypsarrhythmia, as evidenced by EEG findings. Profound hearing loss and brain atrophy were confirmed through MRI imaging. To assess the functional impact of this novel human gene variant, we engineered transgenic *Drosophila* lines expressing both wild type and mutant *SART1* under the control of a UAS promoter.

Our MeDUsA analysis suggested that the variant may confer a gain-of-toxic-function (Nitta et al., 2023). Moreover, we identified heterozygous loss-of-function mutations in *DHX9* as potentially causative for a newly characterized neurodevelopmental disorder. We further investigated the pathogenic potential of a novel heterozygous de novo missense mutation in *DHX9* in a patient presenting with short stature, intellectual disability, and myocardial compaction. Our findings indicated a loss of function in the G414R and R1052Q variants of *DHX9* (Yamada et al., 2023). This experimental framework has been instrumental in elucidating the impact of gene mutations, enhancing our ability to diagnose how novel variants influence gene function.”

Recommendations for the Authors:

Reviewer #1 (Recommendations For The Authors):

Overall I enjoyed reading this paper. It is well presented and represents a significant amount of well executed study. I feel it further characterizes a poorly understood model

of OPA1 variants and one which displays significant differences with the human phenotype. However I feel the use of this model with the author's experiments are not enough to validate this model/experiment as a screening tool for dominant negativity. I have therefore suggested the above experiments as a way to both further validate the mitochondrial dysfunction in this model and to ensure that the expressed transcript is able affect oligomerization as this is a pre-requisite to the authors conclusions.

We assessed the extent to which our model reflects mitochondrial dysfunction using COXII, Atg8, and MitoSOX markers. Unfortunately, neither COXII levels nor the ratio of Atg8a-1 to Atg8a-2 showed significant variations across genotypes that would clarify the impact of dominant negative mutations. Nonetheless, MitoSOX and mito-QC results revealed that mitochondrial ROS levels and mitophagy are increased in *Drosophila* following intrinsic knockdown of dOPA1. These findings are documented in Figures 2, 5, and S6.

Regarding oligomer formation, the specifics remain elusive in this study. However, the expression of *dOPA1K273A*, identified as a dominant negative variant in *Drosophila*, significantly disrupted retinal axon organization, as detailed in Figure S7. From these observations, we hypothesize that oligomerization of wild-type and dominant negative forms in *Drosophila* results in axonal degeneration. Conversely, co-expression of *Drosophila* wild-type with human dominant negative forms does not induce degeneration, suggesting that they likely do not interact.

Reviewer #2 (Recommendations For The Authors):

Materials and Methods:

The authors used GMR-Gal4 to express OPA1-RNAi. I) GMR is expressed in most cells in the developing eye behind the morphogenetic furrow. So the defects observed can be due to knock- down in support cells rather than in photoreceptor cells.

We have added the following sentences in the result (lines 194–196). "The *GMR-Gal4* driver does not exclusively target Gal4 expression to photoreceptor cells. Consequently, the observed retinal axonal degeneration could potentially be secondary to abnormalities in support cells external to the photoreceptors."

OPA1-RNAi: how complete is the knock-down? Have the authors tested more than one RNAi line?

We conducted experiments with an additional RNAi line, and similarly observed degeneration in the retinal axons (Figure S2 A and B; lines 178–179).

The loss-of-function allele, induced by a P-element insertion, gives several eye phenotypes when heterozygous (Yarosh et al., 2008). Does RNAi expression lead to the same phenotypes?

A previous report indicated that the compound eyes of homozygous mutations of *dOPA1* displayed a glossy eye phenotype (Yarosh et al., 2008). Upon knocking down *dOPA1* using the *GMR-Gal4* driver, we also observed a glossy eye-like rough eye phenotype in the compound eyes. These findings have been added to Figure S3 and lines 192–194.

There is no description on the way the somatic clones were generated. How were mutant cells in clones distinguished from wild-type cells (e. g. in Fig. 4).

In the Methods section, we described the procedure for generating clones and their genotypes as follows (lines 502–505): "The *dOPA1* clone analysis was performed by inducing flippase

expression in the eyes using either *ey-Gal4* with *UAS-flp* or *ey3.5-flp*, followed by recombination at the chromosomal location FRT42D to generate a mosaic of cells homozygous for *dOPA1s3475*." Furthermore, we have created a table detailing these genotypes. In these experiments, it was not possible to differentiate between the clone and WT cells. Accordingly, we have noted in the Results section (lines 201–203): "Note that the mutant clone analysis was conducted in a context where mutant and heterozygous cells coexist as a mosaic, and it was not possible to distinguish between them."

Why were flies kept at 29{degree sign}C? this is rather unusual.

Increased temperature was demonstrated to induce elevated expression of GAL4 (Kramer and Staveley, Genet. Mol. Res., 2003), which in turn led to an enhanced expression of the target genes. Therefore, experiments involving knockdown assays or Western blotting to detect human OPA1 protein were exclusively conducted at 29°C. However, all other experiments were performed at 25°C, as described in the methods sections: "Flies were maintained at 25°C on standard fly food. For knockdown experiments (Figures 1C–E, 1F–H, 2A–H, 3B–K, 5F, S1, S2 A and B, and S6A), flies were kept at 29°C in darkness." Furthermore, "We regulated protein expression temporally across the whole body using the *Tub-Gal4* and *Tub-GAL80TS* system. Flies harboring each *hOPA1* variant were maintained at a permissive temperature of 20°C, and upon emergence, females were transferred to a restrictive temperature of 29°C for subsequent experiments."

Legends:

It would be helpful to have a description of the genotypes of the flies used in the different experiments. This could also be included as a table.

We have created a table detailing the genotypes. Additionally, in the legend, we have included a note to consult the supplementary table for genotypes.

Results:

Line 141: It is not clear what they mean by "degradation", is it axonal degeneration? And if so, what is the argument for this here?

In the manuscript, we addressed the potential for mitochondrial degradation; however, recognizing that the expression was ambiguous, the following sentence has been omitted: "Nevertheless, the degradation resulting from mitochondrial fragmentation may have decreased the mitochondrial signal."

Fig. 2: Axons of which photoreceptors are shown?

We have added "a set of the R7/8 retinal axons" to the legend of Figure 2.

Line 167: The authors write that axonal degeneration is more severe after seven days than after eclosion. Is this effect light-dependent? The same question concerns the disappearance of the rhabdomere (Fig. 3G–J).

We conducted the experiments in darkness, ensuring that the observed degeneration is not light-dependent. This condition has been added to the methods section to clarify the experimental conditions.

Line 178/179: Based on what results do they conclude that there is degeneration of the "terminals" of the axons?

Quantification via MeDUSA has enabled us to count the number of axonal terminals, and a noted decrease has led us to conclude axonal terminal degeneration. We have published two papers on these findings. We have added the following description to the results section to clarify how we defined degeneration (lines 174–176): "We have assessed the extent of their reduction from the total axonal terminal count, thereby determining the degree of axonal terminal degeneration (Richard JNS 2022; Nitta HMG 2023).

Line 189: They write: "... we observed dOPA1 mutant axons...". How did they distinguish es mutant from the controls?

Fig. 5 and Fig. 6: How did they distinguish genetically mutant cells from genetically control cells in the somatic clones?

Mutant clone analysis was conducted in a context where mutant and heterozygous cells coexist as a mosaic, and it was not possible to distinguish between them. Accordingly, this point has been added to lines 201–203, "Note that the mutant clone analysis was conducted in a context where mutant and heterozygous cells coexist as a mosaic, and it was not possible to distinguish between them." and the text in the results section has been modified as follows:

(Before "To determine if *dOPA1* is responsible for axon neurodegeneration, we observed the *dOPA1* mutant axons by expressing full-length versions of *dOPA1* in the photoreceptors at one day after eclosion and found that *dOPA1* expression significantly rescued the axonal degeneration" —>

(After "To determine if *dOPA1* is responsible for axon neurodegeneration, we quantify the number of the axons in the *dOPA1* eye clone fly with the expression of *dOPA1* at one day after eclosion and found that *dOPA1* expression partially rescued the axonal degeneration"

Line 225/226: It is not clear to me how their approach "can quantitatively measure the degree of LOF".

To address the reviewer's question and clarify how our approach quantitatively measures the degree of loss of function (LOF), we revised the statement (lines 238–247):

"Our methodology distinctively facilitates the quantitative evaluation of LOF severity by comparing the rescue capabilities of various mutations. Notably, the 2708-2711del and I382M mutations demonstrated only partial rescue, indicative of a hypomorphic effect with residual activity. In contrast, the D438V and R445H mutations failed to show significant rescue, suggesting a more profound LOF. The correlation between the partial rescue by the 2708-2711del and I382M mutations and their classification as hypomorphic is significant. Moreover, the observed differences in rescue efficacy correspond to the clinical severities associated with these mutations, namely in DOA and DOA plus disorders. Thus, our results substantiate the model's ability to quantitatively discriminate among mutations based on their impact on protein functionality, providing an insightful measure of LOF magnitude."

Discussion:

*Line 251, 252 and line 358: What is "the optic nerve" in the adult *Drosophila*?*

In humans, the axons of retinal ganglion cells (RGCs) are referred to as the optic nerve, and we posit that the retinal axons in flies are similar to this structure. In the introduction section, where it is described that the visual systems of flies and humans bear resemblance, we have appended the following definition (lines 107–108): "In this study, we defined the retinal axons of *Drosophila* as analogous to the human optic nerve."

Line 344: These bands appear only upon overexpression of the hOPA1 constructs, so this part of the is very speculative.

Confirmation was achieved using anti-hOPA1, demonstrating that myc is not nonspecific. These results have been added to Figure 5D. Furthermore, the phrase “The upper band was expected as” has been revised to “From a size perspective, the upper band was inferred to represent the full-length hOPA1 including the mitochondria import sequence (MIS).” (lines 464–465)

I was missing a discussion about the increase of ROS upon loss/reduction of dOPA1 observed by others and described here. Is there an increase of ROS upon expression of any of the constructs used?

We demonstrated that not only axonal degeneration but also ROS can be suppressed by expressing human *OPA1* in the genetic background of *dOPA1* knockdown. Additionally, rescue was not possible with any variants except for I382M. Furthermore, we assessed whether there were changes in ROS in the evaluation of dominant negatives, but no significant differences were observed in this experimental system. These findings have been added to the discussion section as follows (lines 318–328). “Our research established that *dOPA1* knockdown precipitates axonal degeneration and elevates ROS signals in retinal axons. Expression of human *OPA1* within this context effectively mitigated both phenomena; it partially reversed axonal degeneration and nearly completely normalized ROS levels. These results imply that factors other than increased ROS may drive the axonal degeneration observed post-knockdown. Furthermore, while differences between the impacts of DN mutations and loss-of-function mutations were evident in axonal degeneration, they were less apparent when using ROS as a biomarker. The extensive use of transgenes in our experiments might have mitigated the knockdown effects. In a systemic *dOPA1* knockdown, assessments of mitochondrial quantity and autophagy activity revealed no significant changes, suggesting that the cellular consequences of reduced *OPA1* expression might vary across different cell types.”

Reviewer #3 (Recommendations For The Authors):

*Consider being more explicit regarding literature that has or has failed to test a direct dominant negative effect by expressing a variant in question in the background of a full *OPA1* complement. My understanding is that this is the first direct evidence of this widely held hypothesis. This lends to the main claim promoting the utility of fly as a model in general. The authors might also outline this in the introduction as a knowledge gap they fill through this study.*

In the introduction, we have incorporated a passage that highlights precedents capable of distinguishing between LOF and DN effects, and we note the absence of models capable of dissecting these distinctions within an *in vivo* organism. This study aims to address this gap, proposing a model that elucidates the differential impacts of LOF and DN within the context of a living model organism, thereby contributing to a deeper understanding of their roles in disease pathology. We added the following sentences in the introduction (lines 71–80).

“In the quest to differentiate between LOF and DN effects within the context of genetic mutations, precedents exist in simpler systems such as yeast and human fibroblasts. These models have provided valuable insights into the conserved functions of *OPA1* across species, as evidenced by studies in yeast models (Del Dotto et al., 2018) and fibroblasts derived from patients harboring *OPA1* mutations (Kane et al., 2017). However, the ability to distinguish between LOF and DN effects in an *in vivo* model organism, particularly at the structural level of retinal axon degeneration, has remained elusive. This gap underscores the necessity for a

more complex model that not only facilitates molecular analysis but also enables the examination of structural changes in axons and mitochondria, akin to those observed in the actual disease state.”

The authors should clarify the language used in the abstract and introduction on the effect of hOPA1 DOA and DOA plus on the dOPA1- phenotype. Currently written as "none of the previously reports mutations known to cause DOA or DOA plus were rescued, their functions seems to be impaired." but presumably the authors mean that these variants failed to rescue to the dOPA1 deficient phenotype.

We thank the reviewer for the constructive feedback. We acknowledge the need for clarity in our description of the effects of *hOPA1* DOA and DOA plus mutations on the *dOPA1*-phenotype in both the abstract and the introduction. The current phrasing, "none of the previously reported mutations known to cause DOA or DOA plus were rescued, their functions seem to be impaired," may indeed be confusing. To address your concern, we have revised this statement to more accurately reflect our findings: "Previously reported mutations failed to rescue the *dOPA1* deficiency phenotype." For Abstract site, we have changed as following. "we could not rescue any previously reported mutations known to cause either DOA or DOA plus." → "mutations previously identified did not ameliorate the *dOPA1* deficiency phenotype."

DOA plus is associated with a multiple sclerosis-like illness; as written it suggests that the pathogenesis of sporadic multiple sclerosis and that associated with DOA plus share and underlying pathogenic mechanism. Please use the qualifier "-like illness."

We have added the term “multiple sclerosis-like illness” wherever “multiple sclerosis” is mentioned.

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